UNIT - I

Engineering Seismology & Theory of Vibrations

Syllabus (as given):

Earthquake phenomenon - causes - faults - plate tectonics - seismic waves - terms - magnitude/intensity - scales - energy - instruments (seismogram, seismoscope, seismograph) - strong ground motion - seismic zones of India. Elements of vibratory system - DOF - continuous system - lumped mass - SHM - free vibration of SDOF (undamped/damped) - critical damping - log decrement - forced vibration - DMF - rigid base excitation - earthquake ground motion.

PART A : ENGINEERING SEISMOLOGY

1.1 Earthquake Phenomenon

Definition - Earthquake

A sudden release of strain energy stored in the earth's crust/upper mantle, which travels as seismic waves and shakes the ground surface.

- *Earth's structure: Crust (5-70 km) -> Mantle (~2900 km thick) -> Outer core (liquid, to ~5150 km) -> Inner core (solid, radius ~1220 km).*
- *Lithosphere (crust + top mantle, ~100 km) is rigid and broken into plates; below it the soft asthenosphere flows slowly (convection currents).*
- *Strain builds up for years at plate edges/faults; when rock strength is exceeded it ruptures and slips -> earthquake.*
- *Most earthquakes (~90%) are tectonic and occur at depths < 70 km (shallow), hence very destructive.*

Term	Meaning
Focus / Hypocentre	Point inside earth where rupture starts and energy is released
Epicentre	Point on ground surface vertically above the focus
Focal depth	Vertical distance focus -> epicentre (shallow < 70 km, intermediate 70-300 km, deep > 300 km)
Epicentral distance	Distance from the observation station to epicentre
Hypocentral distance	Distance from station to focus
Foreshock / Aftershock	Smaller shocks before / after the main shock in the same region
Isoseismals	Lines joining places of equal intensity on a map
Meizoseismal area	Area of maximum damage, surrounding the epicentre
Seismic gap	Active fault segment that has not ruptured for long: likely site of future shock

Terms associated with earthquakes

1.2 Causes of Earthquakes

- Tectonic (most important): sudden slip along faults / plate boundaries - Elastic Rebound Theory (Reid, 1906, after the 1906 San Francisco EQ).
- Volcanic: movement of magma and gases; usually small, local.
- Induced / man-made: reservoir-induced seismicity (Koyna 1967, M 6.5), mining, deep well injection, nuclear blasts.
- Others: landslides, rock-falls, meteorite impact, collapse of caverns.

Important / Exam point

Elastic Rebound Theory: rocks on either side of a fault are slowly strained -> strain exceeds frictional strength -> sudden slip -> rocks "rebound" to a less-strained position and the stored elastic energy is released as waves.

1.3 Faults

Definition - Fault

A fracture/zone in the rock across which there has been relative displacement. Dip = angle of fault plane from horizontal; Strike = direction of fault line on the surface.

Type	Movement	Stress / Notes
Normal (gravity) fault	Hanging wall moves DOWN relative to footwall	Tension; crust extends (divergent zones)
Reverse / Thrust fault	Hanging wall moves UP over footwall	Compression; crust shortens (Himalaya). Thrust = low dip (<45°)
Strike-slip fault	Horizontal sliding, along the strike	Shear; right-lateral or left-lateral (San Andreas)
Oblique-slip fault	Combination of dip-slip and strike-slip	Most real faults

1.4 Plate Tectonics

Earth's lithosphere is divided into 7 major plates (Pacific, Eurasian, African, North American, South American, Indo-Australian, Antarctic) and several minor ones. Plates move 2-15 cm/year driven by mantle convection, ridge-push and slab-pull.

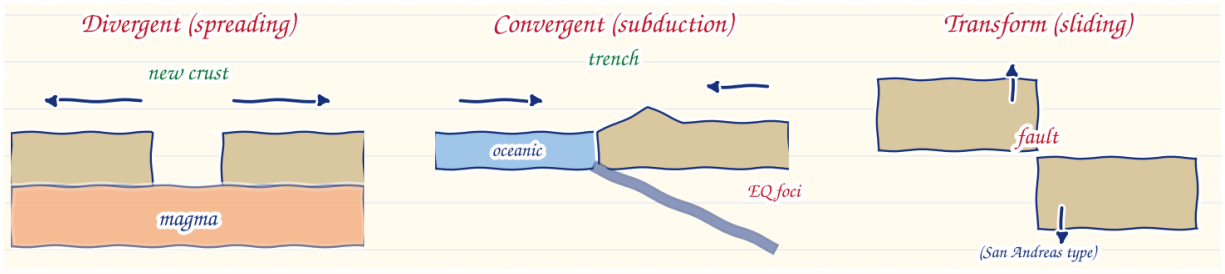


Fig: Types of plate boundaries

- Divergent boundary - plates move apart; new crust forms (Mid-Atlantic ridge); shallow, small EQs.
- Convergent boundary - plates collide: ocean-continent (subduction, deep EQs, volcano arcs - Andes), ocean-ocean (island arcs - Japan), continent-continent (mountain building - Himalaya:

Indian plate moves ~5 cm/yr north under Eurasia).

- *Transform boundary* - plates slide past; no crust created or destroyed; shallow strong EQs (San Andreas).
- *Intra-plate earthquakes* occur inside plates at old weak zones, e.g. Latur 1993, Bhuj 2001, Koyna - India's peninsular region.

> *Important / Exam point*
> India: the Indian plate collides with Eurasia -> Himalayan belt is among the most seismically active zones; 1897 Shillong, 1905 Kangra, 1934 Bihar-Nepal, 1950 Assam, 2001 Bhuj, 2015 Nepal.

1.5 Seismic Waves

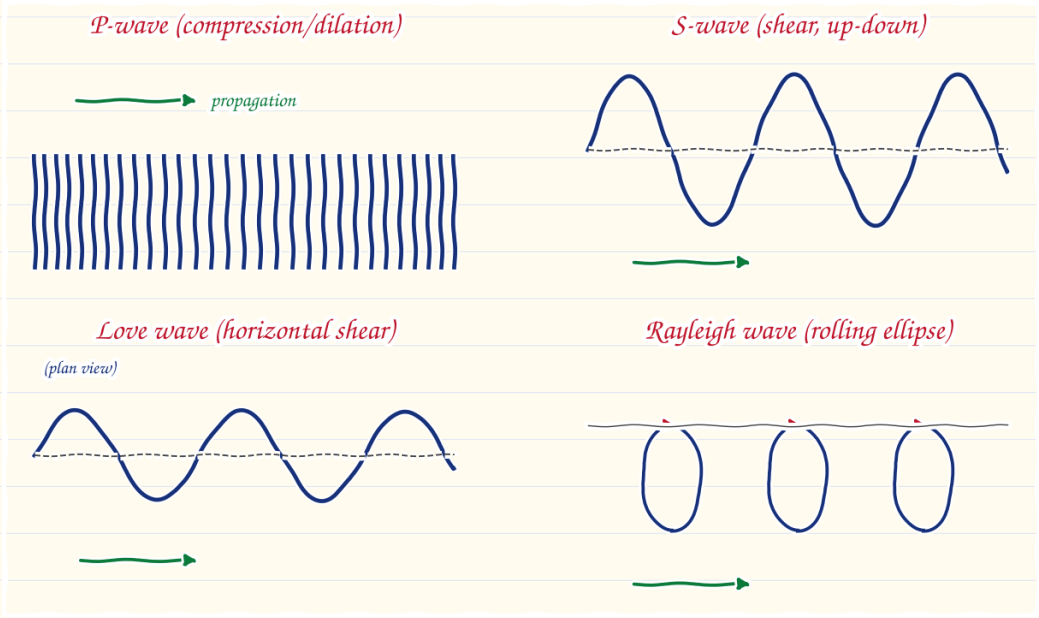


Fig: Seismic wave types

Wave	Class	Motion	Speed / Remarks
P (Primary)	Body, fastest	Particle motion parallel to travel (push-pull, compression)	$V_p = \sqrt{[(K+4\mu/3)/\rho]}$ ~ 5-8 km/s; travels through solid and liquid; first arrival
S (Secondary)	Body	Perpendicular to travel (shear)	$V_s = \sqrt{(\mu/\rho)}$ ~ 3-4.5 km/s; only solids; arrives second; causes most structural damage
Love (L)	Surface	Horizontal shear, perpendicular to travel	Slightly slower than S; very damaging to foundations
Rayleigh (R)	Surface	Retrograde elliptical in vertical plane (like ocean waves)	Slowest (~0.9 V_s); large amplitude; long period

- Order of arrival: P -> S -> Love -> Rayleigh. S-P time lag gives epicentral distance (about 8 km per second of lag).
- Surface waves decay slowly with distance, body waves faster. Surface waves dominate far-field and long-period motion (tall buildings, bridges).

- S-waves not passing through the outer core proves that the outer core is liquid (shadow zone).

1.6 Magnitude and Intensity

Definition - Magnitude (M)

A quantitative, instrumental measure of the energy released at the source. One value per earthquake, independent of location.

Richter local magnitude (1935)

$M_L = \log_{10} A - \log_{10} A_0(\delta)$

A = max trace amplitude (mm) on a Wood-Anderson seismograph at 100 km; A_0 = standard amplitude.

Other scales: body-wave m_b, surface-wave M_s, moment magnitude $M_w = (2/3) \log_{10} M_0 - 10.7$ ($M_0 = \mu A D$ in dyne-cm; does not saturate - preferred today).

Energy released (Gutenberg-Richter)

$\log_{10} E = 1.5 M + 4.8$ (E in joules)

Each unit increase in M -> amplitude x10, energy x 31.6 (~32). Two units -> energy x 1000.

e.g. M 7 releases 32 times the energy of M 6.

Definition - Intensity (I)

A qualitative measure of effects/damage at a given place (felt by people, damage to structures, ground changes). Differs from place to place; highest near epicentre.

Scale	Range	Remarks
Modified Mercalli (MMI)	I - XII	Most used internationally: I = not felt ... VII = damage to poorly built structures ... XII = total destruction
Rossi-Forel	I - X	Older scale
MSK-64 (Medvedev-Sponheuer-Karnik)	I - XII	Used in India (IS 1893 uses MSK); defines building types A, B, C
EMS-98 (European Macroseismic)	I - XII	Modern, vulnerability-class based
Richter / Moment	0 - ~9.5 (open ended)	Magnitude scales, log based

Magnitude vs Intensity	
Magnitude	Intensity
Energy at source; single number	Effect at a location; varies with distance, soil, building quality
Measured by seismograph	Assessed from observations / questionnaires
Arabic numerals (6.5)	Roman numerals (VII)

1.7 Earthquake Measuring Instruments

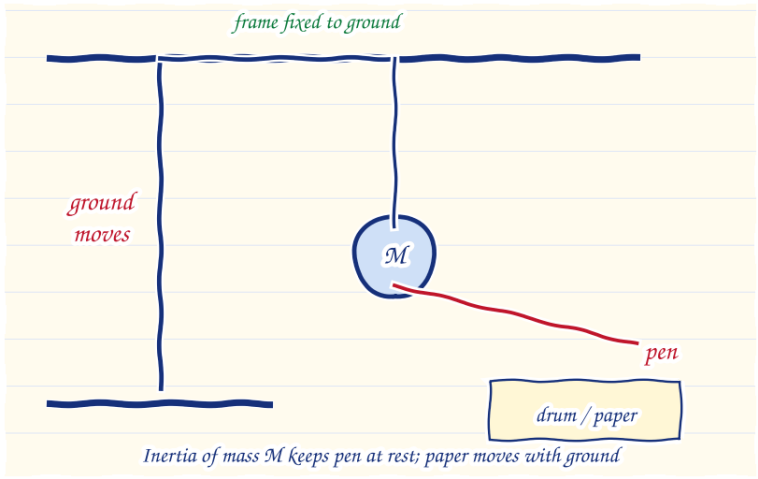


Fig: Principle of a seismograph: heavy mass stays still, frame moves

- Seismoscope - simple device that only indicates that a shock occurred (and maybe its direction/rough amplitude); no time scale, no true record.
- Seismograph - records ground motion vs time with a mass-spring-damper (inertial) sensor, amplifier and recorder; types: short-period, long-period, broadband; horizontal and vertical components. Natural period of sensor is selected so that ground displacement ($T_n >> T$) or ground acceleration ($T_n << T$) is recorded.
- Seismogram - the record produced (trace of ground displacement/velocity/acceleration vs time). P, S and surface wave arrivals are visible; S-P interval -> epicentral distance; amplitude -> magnitude.
- Accelerograph (strong-motion instrument) - records ground acceleration at strong shaking without going off-scale (SMA-1 in India); the record is an accelerogram giving PGA, frequency content, duration. Three components (2 horizontal + 1 vertical).
- Modern: digital broadband sensors, GPS, networks of the National Seismological Network (NCS, IMD); epicentre located by triangulation from at least 3 stations.

1.8 Strong Ground Motion

Strong ground motion is the shaking strong enough to affect structures. Its engineering characteristics:

- Amplitude: Peak Ground Acceleration (PGA), velocity (PGV) and displacement (PGD). Zone V design PGA ~ 0.18 g (DBE) / 0.36 g (MCE).
- Frequency content: dominant periods (response spectrum / Fourier spectrum). Hard rock -> high frequency; soft soil -> long periods, amplified.
- Duration: bracketed or significant duration - longer shaking = more damage accumulation (low-cycle fatigue).
- Factors: magnitude, source mechanism, distance, path (attenuation), local soil conditions (site amplification, liquefaction), directivity, near-fault (fling/pulse) effects.
- Typical accelerograms: 1940 El Centro (PGA 0.32 g), 1995 Kobe, 2001 Bhuj.

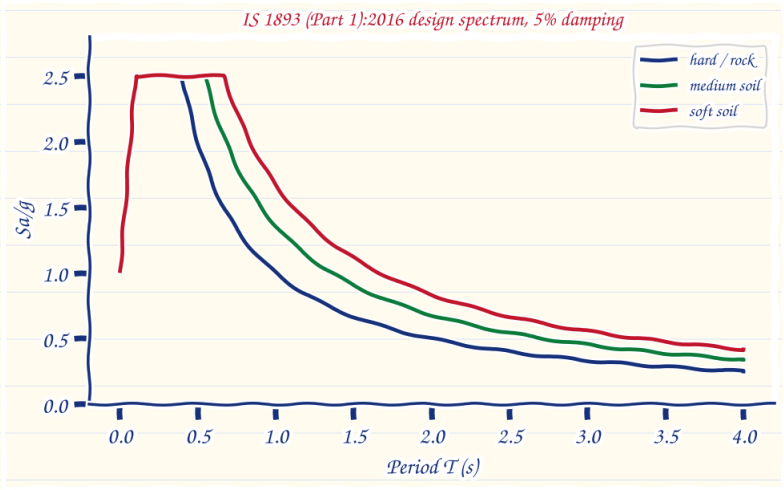


Fig: IS 1893 (Part 1):2016 spectra for different soils

1.9 Seismic Zones of India (IS 1893 (Part 1):2016)

The 2016 revision merges the old Zone I with Zone II -> only four zones: II, III, IV and V (earlier code 2002 had Zone I-V; old Zone I removed). Map is based on maximum considered earthquake (MCE) intensity and past seismicity.

Zone	Seismic intensity	Zone factor Z (MCE)	Design Z/2 (DBE)	Areas (examples)
II	Low (MSK VI)	0.10	0.05	Parts of south India: Bengaluru-Tamil Nadu interior, Rajasthan west
III	Moderate (VII)	0.16	0.08	Chennai, Mumbai, Kolkata, Hyderabad, Kerala, much of A.P.
IV	Severe (VIII)	0.24	0.12	Delhi, Patna, Srinagar region, Sikkim, parts of Gujarat & Bihar
V	Very severe (IX +)	0.36	0.18	Kashmir, Himachal W., Uttarakhand, N-E states, Rann of Kutch, Andaman & Nicobar

Important / Exam point

Remember Z values 0.10, 0.16, 0.24, 0.36 -> Z/2 is the DBE (design basis earthquake) factor used in A_h. About 59% of India's land area is prone to moderate-severe shaking; ~11% falls in Zone V.

- *Zone factor Z = effective peak ground acceleration (as fraction of g) for MCE on rock/firm ground.*
- *Seismic microzonation (soil, liquefaction, slope) refines the national map for cities.*

PART B : THEORY OF VIBRATIONS

2.1 Elements of a Vibratory System

- *Mass (m) - stores kinetic energy; inertia force = m ü.*

- *Spring / stiffness (k)* - stores strain energy; restoring force = $k\,u$. For a cantilever column $k = 3EI/h^3$; for a fixed-fixed column $k = 12EI/h^3$.
- *Damper (c)* - dissipates energy; viscous damping force = $c\,\dot{u}$ (proportional to velocity).
- *Excitation* - external force $p(t)$ or ground motion $\ddot{u}_g(t)$.

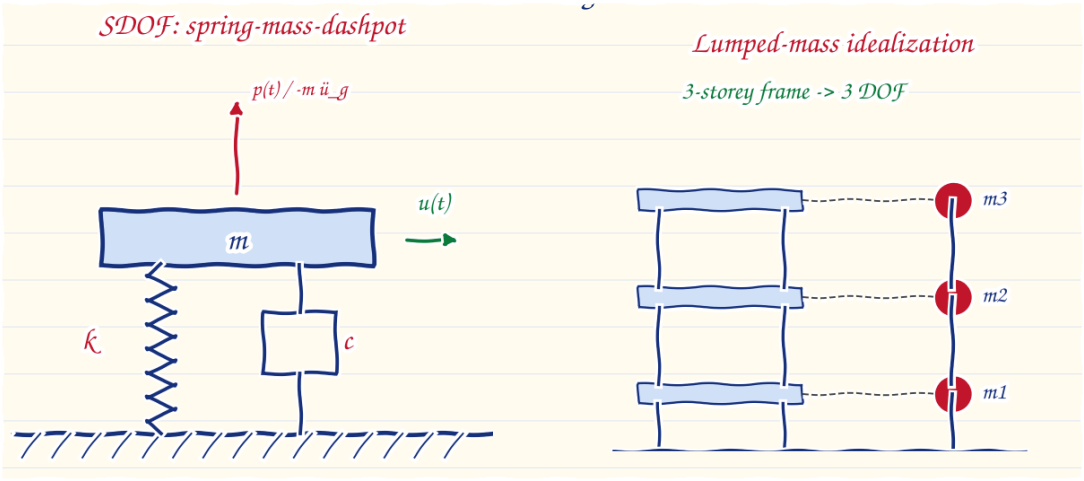


Fig: SDOF model and lumped mass idealization

2.2 Degrees of Freedom and Idealization

> **Definition - Degree of Freedom (DOF)**
> The *minimum number of independent coordinates* needed to completely define the position of all masses of a system at any instant.

- *SDOF* - one coordinate (water tank on a single column, single-storey frame with rigid girder).
- *MDOF* - n coordinates for an n-storey shear building (n lumped masses, each with one lateral translation) -> n natural frequencies.
- *Continuous system* - mass and elasticity distributed (beams, chimneys, dams); *infinite DOF*, governed by partial differential equations.
- *Lumped mass idealization* - replace distributed mass by discrete masses at floor levels (floor mass includes slab, beams, half of columns above and below). Assumptions: floor slabs rigid in-plane, columns massless & axially rigid -> *shear building*; it reduces a continuous problem to a finite number of DOF.

Equation of motion for an n-DOF shear building: $[M]\{\ddot{u}\} + [C]\{\dot{u}\} + [K]\{u\} = -[M]\{1\}\ddot{u}_g$ (K is tridiagonal).

2.3 Oscillatory Motion and Simple Harmonic Motion

- *Periodic motion* repeats after equal time intervals; *harmonic* motion is sinusoidal: $u = U \sin(\omega t + \varphi)$.
- *Simple Harmonic Motion (SHM)*: acceleration proportional to displacement and directed to the mean position: $\ddot{u} = -\omega^2 u$.
- Amplitude U, circular frequency ω (rad/s), cyclic frequency $f = \omega/2\pi$ (Hz), period $T = 1/f = 2\pi/\omega$ (s), phase angle φ.

- Velocity leads displacement by 90°, acceleration by 180° (so max velocity = ωU, max acceleration = $\omega^2 U$).
- Complex form: $u = Re[U e^{i\omega t}]$; two harmonics of same frequency add as vectors (phasors).

2.4 Free Vibration of an SDOF System

(a) Undamped

Equation of motion (Newton / D'Alembert)

$m\ddot{u} + ku = 0$

$\omega_n = \sqrt{(k/m)}$ (rad/s), $T_n = 2\pi/\omega_n = 2\pi\sqrt{(m/k)}$, $f_n = 1/T_n$

Solution: $u(t) = u_0 \cos \omega_n t + (\dot{u}_0/\omega_n) \sin \omega_n t$ (constant amplitude, no decay)

Amplitude $U = \sqrt{[u_0^2 + (\dot{u}_0/\omega_n)^2]}$

Natural period depends only on m and k (not on the initial conditions). Stiffer/lighter -> shorter period. Rough building period: $T \approx 0.1\,N$ seconds (N = number of storeys).

(b) Damped (viscous)

Equation of motion

$m\ddot{u} + c\dot{u} + ku = 0 \rightarrow \ddot{u} + 2\zeta\omega_n \dot{u} + \omega_n^2 u = 0$

Critical damping $c_{cr} = 2\sqrt{(km)} = 2m\omega_n$ Damping ratio $\zeta = c / c_{cr}$

Characteristic roots: $s = -\zeta\omega_n \pm \omega_n \sqrt{(\zeta^2 - 1)}$

Case	Condition	Behaviour
Under-damped	$\zeta < 1$ (all real buildings)	Decaying oscillation: $u = e^{-\zeta\omega_n t}[u_0 \cos \omega_d t + ((\dot{u}_0 + \zeta\omega_n u_0)/\omega_d) \sin \omega_d t]$, damped frequency $\omega_d = \omega_n \sqrt{(1-\zeta^2)}$, $T_d = 2\pi/\omega_d$
Critically damped	$\zeta = 1$ ($c = c_{cr}$)	No oscillation; returns to rest in minimum time: $u = (A + Bt) e^{-\omega_n t}$
Over-damped	$\zeta > 1$	Sluggish non-oscillatory return, slower than critical

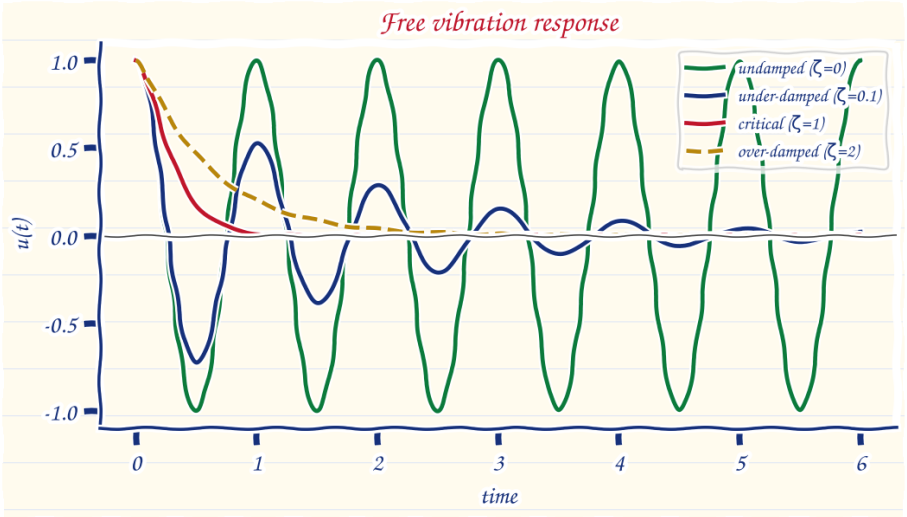


Fig: Free vibration for different damping ratios

- Typical ζ: steel buildings 2-5%, RC buildings 5% (IS 1893 spectra are for 5% damping), masonry 5-7%, bridges 2-5%, structure with dampers 10-30%.
- For $\zeta \le 20\%$, $\omega_d \approx \omega_n$ (difference < 2%). Damping reduces amplitude, not period (practically).

2.5 Logarithmic Decrement

Definition - Logarithmic decrement (δ)
Natural logarithm of the ratio of any two successive peak amplitudes in free damped vibration. It measures the damping experimentally.

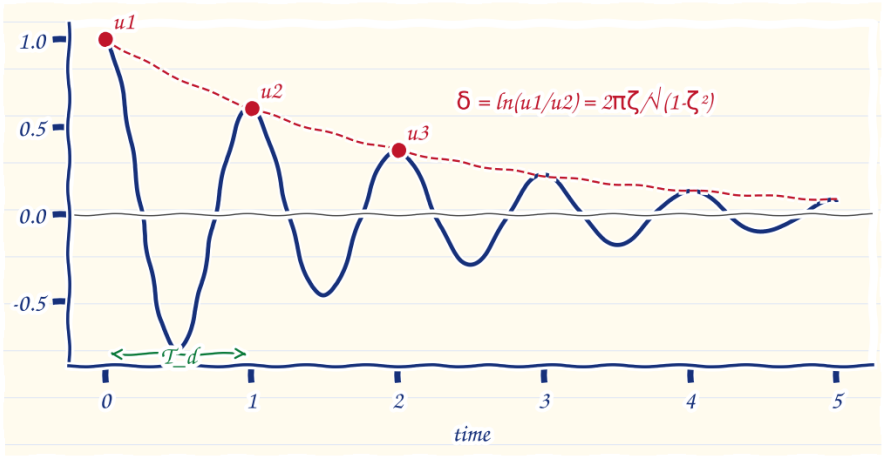


Fig: Decay of free vibration - peaks u1, u2, u3

Formulae
$\delta = \ln(u_1/u_2) = \zeta \omega_n T_d = 2\pi\zeta / \sqrt{(1 - \zeta^2)} \approx 2\pi\zeta$ (for small ζ)
For n cycles apart: $\delta = (1/n) \ln(u_1/u_{n+1})$ -> $\zeta \approx \delta/2\pi$
Amplitude after n cycles: $u_{n+1} = u_1 e^{-n\delta}$

Important / Exam point
Numerical (typical): Peak amplitude falls from 20 mm to 5 mm in 5 cycles -> $\delta = (1/5) \ln(20/5) = 0.277$ -> $\zeta = 0.277/2\pi = 0.044 \approx 4.4\%$.

2.6 Forced Vibration - Harmonic Excitation

Equation: $m\ddot{u} + c\dot{u} + ku = p_0 \sin \omega t$
Total response = complementary (transient, dies out due to damping) + particular (steady state).
Steady state: $u(t) = \rho \sin(\omega t - \varphi)$, static deflection $u_{st} = p_0/k$
$\rho = u_{st} \cdot D$, $D = 1 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$, $r = \omega/\omega_n$
Phase lag $\varphi = \tan^{-1}[2\zeta r / (1 - r^2)]$

Definition - Dynamic Magnification Factor (DMF / D)
Ratio of the dynamic amplitude to the static deflection under the same force amplitude: $D = \rho/u_{st}$.

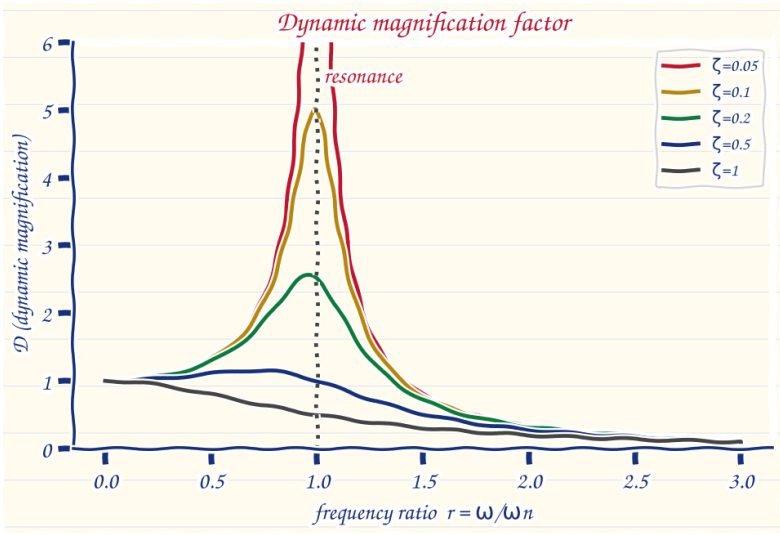


Fig: DMF vs frequency ratio

- $r << 1$ (slow load): $D \approx 1$ - behaves statically (stiffness controlled).
- $r = 1$ (resonance): $D = 1/2\zeta$ ($max_r \approx 1/2\zeta$; infinite if undamped) - damping controlled; $\varphi = 90°$. For $\zeta = 5\%$ $D = 10$!
- $r >> 1$ (fast load): $D \to 0$ - mass (inertia) controlled; structure hardly responds; basis of base isolation.
- Peak of D occurs at $r = \sqrt{(1 - 2\zeta^2)}$ with $D_{max} = 1/(2\zeta\sqrt{(1 - \zeta^2)})$.
- Resonance danger: when ground motion's dominant period ≈ building period (Mexico City 1985: soft lake bed $T \approx 2$ s vs 6-15 storey buildings).

Transmissibility and force transmitted

Force transmitted to support / isolation: $TR = \sqrt{[1 + (2\zeta r)^2]} / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$; isolation effective only for $r > \sqrt{2}$.

2.7 Excitation by Rigid Base Translation (Earthquake)

For ground displacement $u_g(t)$, with u = relative displacement of mass w.r.t. base, total displacement $u_t = u + u_g$:

Equation of motion for ground motion

$m\ddot{u} + c\dot{u} + ku = -m\ddot{u}_g(t) \to \ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2 u = -\ddot{u}_g(t)$

Effective earthquake force $p_{eff}(t) = -m\ddot{u}_g(t)$ (inertia force from base acceleration)

For harmonic base motion $\ddot{u}_g = a \sin \omega t$: relative amplitude $u_0 = (m a/k) D = (a/\omega_n^2) D$

- Only ground acceleration and system properties (ω_n, ζ) control relative response; mass cancels out: response depends on T_n, ζ only.
- Shear force in the spring (column) $V = ku = m\omega_n^2 u \to$ pseudo-acceleration $A = \omega_n^2 u$ gives equivalent static force $f_s = mA$.

2.8 Earthquake Ground Motion - Response to General Ground Acceleration

Duhamel integral (damped SDOF)

$u(t) = -(1/\omega_d) \int_0^t \ddot{u}_g(\tau)\, e^{-\zeta\omega_n(t-\tau)} \sin \omega_d(t-\tau)\, d\tau$

- The accelerogram is applied to many SDOF oscillators with different T_n (same ζ); the maximum response is plotted against T_n -> response spectrum: $S_d = \max|u|$, $S_v = \omega_n S_d$ (pseudo-velocity), $S_a = \omega_n^2 S_d$ (pseudo-acceleration).
- Design Spectrum (smoothed, IS 1893 S_d/g vs T) is the envelope/mean of many spectra for a soil class - used for code analysis.
- Base shear for SDOF: $V_b = m S_a = (S_d/g) W$.
- Earthquake ground motion is random, transient, broad-band with 3 translational components; vertical usually 2/3 of horizontal.

SOLVED EXAMPLES (typical exam numericals)

Ex.1 Natural period of SDOF

Water tank: $m = 20\,000$ kg on column of lateral stiffness $k = 8 \times 10^6$ N/m.

$\omega_n = \sqrt{(k/m)} = \sqrt{(8e6/20000)} = \sqrt{400} = 20$ rad/s

$T_n = 2\pi/\omega_n = 0.314$ s, $f_n = 1/T_n = 3.18$ Hz

Ex.2 Damping ratio and log decrement

Same tank has dashpot $c = 40\,000$ N.s/m.

$c_{cr} = 2\sqrt{(km)} = 2\sqrt{(8e6 \times 20000)} = 8 \times 10^5$ N.s/m -> $\zeta = 40000/800000 = 0.05$ (5%)

$\omega_d = 20\sqrt{(1 - 0.05^2)} = 19.97$ rad/s (almost $= \omega_n$)

$\delta = 2\pi(0.05)/\sqrt{(1-0.05^2)} = 0.3145$ -> $u_1/u_2 = e^{0.3145} = 1.37$ (each peak is 73% of the previous one)

Ex.3 Dynamic magnification

$\zeta = 0.05$, $r = \omega/\omega_n = 0.8$: $D = 1/\sqrt{[(1-0.64)^2 + (2 \times 0.05 \times 0.8)^2]} = 1/\sqrt{(0.1296 + 0.0064)} = 2.71$

At resonance $r = 1$: $D = 1/(2\zeta) = 10$ -> a harmonic force of 10 kN produces 10 times the static deflection (10/k).

Ex.4 Energy and epicentral distance

M 6 -> M 8: $E_8/E_6 = 10^{1.5 \times 2} = 1000$ times.

S-P lag $\Delta t = 20$ s, $V_p = 8$ km/s, $V_s = 4.6$ km/s: $d = \Delta t \times V_p V_s/(V_p - V_s) = 20 \times (8 \times 4.6)/3.4 =$ ~216 km.

Formula sheet

Short-answer table: $\omega_n = \sqrt{(k/m)}$; $c_{cr} = 2\sqrt{(km)}$; $\zeta = c/c_{cr}$; $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$; $\delta \approx 2\pi\zeta$; $D_{res} = 1/2\zeta$; $Sa = \omega_n^2 Sd$; $V_b = (Sa/g)W$.

QUICK REVISION - Likely questions

1. Explain elastic rebound theory and types of faults with sketches.

2. Describe plate tectonics and the types of plate boundaries; why does India experience earthquakes?
3. Classify seismic waves; derive/state V_p and V_s; sequence of arrival.
4. Differentiate magnitude and intensity; Richter scale, energy relation $\log E = 1.5M + 4.8$.
5. Seismograph vs seismoscope vs accelerograph; characteristics of strong ground motion.
6. Seismic zones of India as per IS 1893 (Part 1):2016 with Z values.
7. Derive free vibration response of damped SDOF; define critical damping; log decrement.
8. Define DMF; derive expression; effect of damping near resonance.
9. Derive equation of motion of SDOF subjected to ground motion.

UNIT - II

Conceptual Design & Introduction to Earthquake Resistant Design

Syllabus (as given):

Functional planning - continuous load path - overall form - simplicity & symmetry - elongated shapes - stiffness & strength - horizontal & vertical members - twisting - ductility - flexible buildings - framing systems - choice of materials (unconfined/confined concrete, masonry, steel). Seismic design requirements - regular/irregular configurations - assumptions - design loads - load combinations - permissible stresses - methods of analysis - factors in analysis - equivalent lateral force method.

PART A : CONCEPTUAL DESIGN

1.1 Introduction

Conceptual (architectural + structural) design decides how a building will behave in an earthquake before any calculation is done. Good configuration is cheaper and more reliable than heavy detailing on a poor configuration. "A good building is made, not analysed" - failures in Bhuj 2001, Kobe 1995, Northridge 1994 were mostly due to bad configuration: soft storeys, torsion, short columns, pounding.

Definition - Seismic design philosophy (IS 1893)

Structure should: (i) resist minor shaking (frequent) without damage; (ii) resist moderate shaking with limited structural damage and some non-structural damage (repairable); (iii) resist major (MCE) shaking without collapse - life safety, with damage allowed. Achieved by strength + ductility (reducing the design force by factor R).

- Why not design elastic for MCE? Elastic design force would be R times larger (3-5x) -> uneconomical. Instead allow inelastic energy dissipation.
- Four virtues of EQ-resistant buildings (Arnold & Reitherman): Good structural configuration, adequate lateral stiffness, adequate lateral strength, good ductility.
- Relevant codes: IS 1893 (Part 1):2016 (design loads), IS 13920:2016 (ductile detailing of RC), IS 4326:2013 (low-rise buildings), IS 13828 (low-strength masonry), IS 13827 (earthen), IS 15988 (retrofit of RC).

1.2 Functional Planning

- Plan the layout of rooms, stairs, lifts, services so that structural elements can be arranged symmetrically and uniformly - do not let architecture demand floating columns, large openings in floors, discontinuous walls.
- Locate heavy masses (water tanks, storage, plant rooms) low and near the centre; avoid heavy roof loads (reduces base shear since $V = A_h W$).

- Stair/lift cores should be symmetrically located or isolated from the main frame (otherwise they stiffen one side -> torsion).
- Provide seismic joints (separation gaps) where a building has sudden changes in shape, so each part vibrates independently; also to avoid pounding with adjacent buildings.
- Keep ground storey open car-parking (stilt) only if the frame is specially stiffened - otherwise it becomes a soft storey.
- Keep spans moderate, storey heights uniform, light non-structural partitions.

> Seismic gap
> Gap for pounding (IS 1893:2016 cl. 7.11.1.3): separation = R x design lateral displacement of each adjacent unit (sum of the two); typically 25-40 mm for low-rise. Floors of adjacent buildings should ideally be at the same level.

1.3 Continuous Load Path

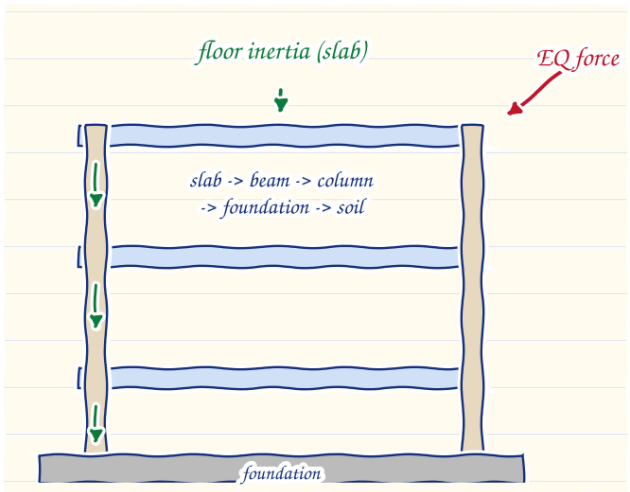


Fig: Load path for gravity and lateral (earthquake) forces

- A continuous, direct load path must carry inertia forces from every mass (slab, walls, roof) -> diaphragm -> vertical elements (columns/shear walls) -> foundation -> soil. Any break (weak link) is where failure starts.
- Floor slabs act as rigid diaphragms distributing lateral loads to vertical resisting elements by stiffness.
- Connections: beam-column joints, roof-to-wall, wall-to-foundation, wall-to-wall (corner) must be at least as strong as members connected.
- Walls/columns should run vertically down to the foundation without discontinuity (avoid floating columns, hanging walls, discontinuous shear walls).

1.4 Overall Form - Simplicity and Symmetry

- Simple, compact, symmetrical plan in both axes (square, rectangle, circle) gives uniform stress, minimal torsion, easy to analyse and detail. Symmetry applies to both geometry (mass) and stiffness.
- Centre of mass (CM) and centre of rigidity/stiffness (CR) should coincide: eccentricity e = |CM - CR| as small as possible. Torsional moment = V x e.

- Complex shapes (L, T, U, H, Y, +, E) develop stress concentration at re-entrant corners and different wings vibrate with different periods -> separate them by seismic joints into simple rectangles.
- Vertical form: continuous, uniform elevation; avoid abrupt set-backs, overhangs (cantilevers), heavy upper floors, tapered mass at top.

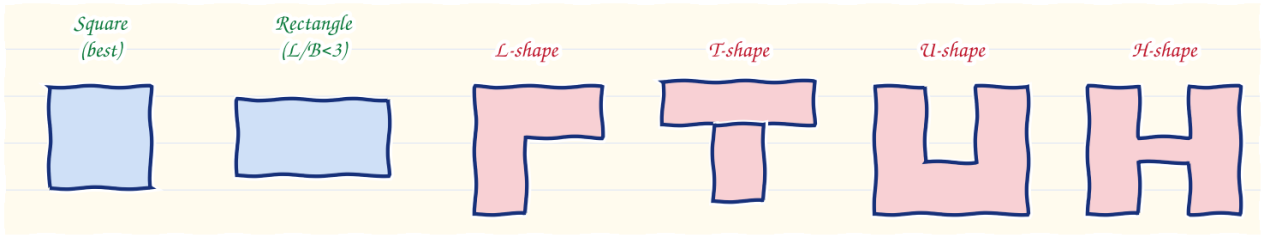


Fig: Regular (blue/good) vs irregular plan shapes (pink/poor)

Elongated shapes

- Long buildings (L/B > 3) suffer because ground motion arrives with phase difference at the ends; long floor diaphragms behave flexibly and are subjected to differential motion -> tension/cracking in the diaphragm, large torsion.
- Limit plan aspect ratio L/B ≤ 3 (IS 4326 limits to ≤ 3 for masonry); otherwise provide seismic joints at suitable intervals.
- Narrow tall buildings (H/B > 4) -> overturning, large axial force changes in end columns; keep slenderness H/B small.

1.5 Stiffness and Strength; Horizontal & Vertical Members

- Stiffness (lateral) controls period T, storey drift and non-structural damage. Stiffness too small -> large drift (P-delta, damage); too large -> attracts higher force (A_h increases as T decreases up to the plateau).
- Strength controls whether the structure yields. Strength must be uniform in plan and elevation - avoid weak storey (strength < 80% of storey above).
- Vertical members (columns, shear walls) carry gravity and resist lateral loads; they must be strong and ductile - "strong column - weak beam" so that yielding occurs in beams (energy dissipated in many places) and not in columns (storey collapse).
- Horizontal members (beams, slabs/diaphragms) transfer inertia forces to vertical members; beam failure is less catastrophic than column failure. Keep slab continuous (opening < 50% of area), well connected to walls/columns.
- Use shear walls/bracing at periphery to give maximum torsional resistance.

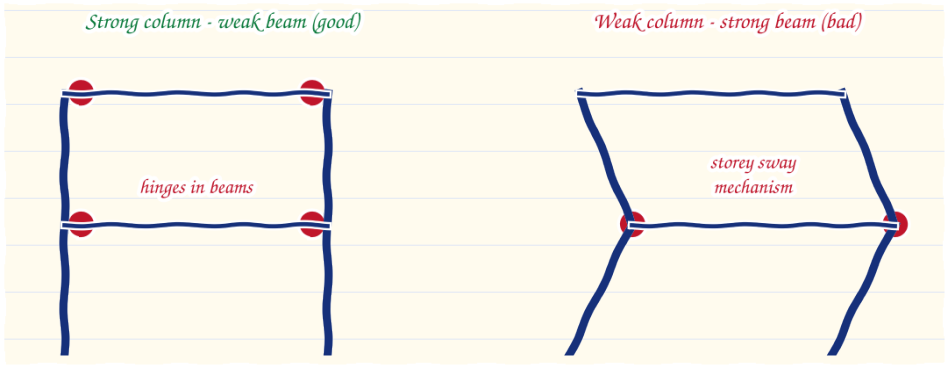


Fig: Beam-sway vs storey mechanism

1.6 Twisting of Buildings (Torsion)

- Arises when CM ≠ CR (mass or stiffness unsymmetrical), or from rotational component of ground motion -> building rotates about CR, columns far from CR undergo extra displacement.
- Torsion causes high demand in corner/edge columns, large differential drift and often collapse (e.g., corner buildings with ground floor stiffness only on two sides).
- IS 1893: design eccentricity $e_d = 1.5\, e_s + 0.05\, b$ or $e_s - 0.05\, b$ (more severe) where e_s = static eccentricity, b = plan dimension perpendicular to force.
- Remedies: symmetric layout; place stiff elements (walls, cores) at the periphery, symmetrically; avoid core placed at one end; increase torsional stiffness; keep mass symmetric.
- Torsional irregularity: max storey drift at one end > 1.5 x min drift (at the other end) (IS 1893:2016).

1.7 Ductility

Definition - Ductility (μ)
The ability of a structure/member to undergo large inelastic deformation without significant loss of strength, thereby dissipating energy. Displacement ductility $\mu = \Delta_u / \Delta_y$.

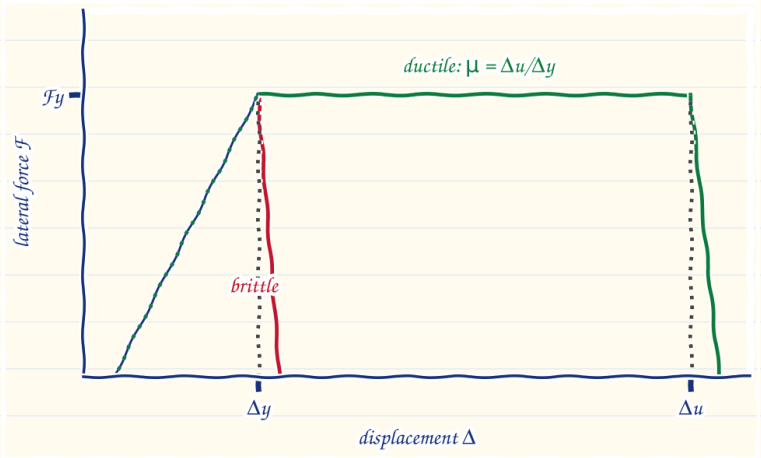


Fig: Ductile vs brittle response

- Ductility relationships: displacement $\mu_\Delta = \Delta_u / \Delta_y$; curvature $\mu_\varphi = \varphi_u / \varphi_y$; rotational $\mu_\theta = \theta_u / \theta_y$; Relationship: $\mu_\Delta = 1 + 3(\mu_\varphi - 1)(L_p/L)(1 - 0.5\, L_p/L)$ (L_p = plastic hinge length).
- Response reduction R and ductility: equal displacement rule (long period): $R_\mu = \mu$; equal energy rule (short period, T < ~0.5 s): $R_\mu = \sqrt{(2\mu - 1)}$.

- Ductility is provided by: closely spaced ties/confinement, adequate anchorage, controlled steel ratio (under-reinforced), tension-controlled sections, good detailing (IS 13920).
- Brittle failures to avoid: shear failure, bond/anchorage failure, crushing of concrete, buckling of bars, joint failure, masonry diagonal tension.
- Capacity design: design the intended yielding parts (beams) first, then make the rest (columns, joints, shear capacity) stronger so that ductile mechanism forms.

1.8 Flexible Buildings

- Flexible (tall, slender, moment frame) buildings have long natural period ($T \approx 1$-4 s) -> attract lower design seismic force (S_d/g falls with T) but suffer larger displacements and drift.
- Problems: large inter-storey drift damaging partitions and cladding, P-delta effects, instability, pounding with neighbours; resonance with long-period ground motion on soft soils.
- Stiff (low-rise, walled) buildings: short period, higher force, small drift, but less ductile and may be brittle (masonry).
- Balanced design: limit drift to 0.004 h under design EQ (IS 1893:2016), keep adequate stiffness using shear walls, bracing; $T \leq 4$ s ideal.

1.9 Framing Systems

System	Features	Remarks
Moment-resisting frame (MRF)	Beams & columns with rigid joints resist lateral loads by flexure	Architecturally flexible; ductile if SMRF ($R = 5$); drift is main issue; best up to ~10-15 storeys
Braced frame (concentric/eccentric)	Diagonal members carry lateral load axially (steel)	Very stiff; EBF more ductile (link beams)
Shear wall system	RC walls resist lateral loads by cantilever action	Stiff and strong; reduces drift; ductile shear wall $R = 4$
Dual system (wall-frame)	Frame + walls share lateral load (frame ≥ 25% of base shear)	Best performance for 10-40 storeys; R = 4.5-5
Tube / core-outrigger	Perimeter closely spaced columns acting as tube	Tall buildings > 40 storeys
Infilled frame	Masonry infill within frame	Stiffens frame; but short-column/soft-storey risk, brittle

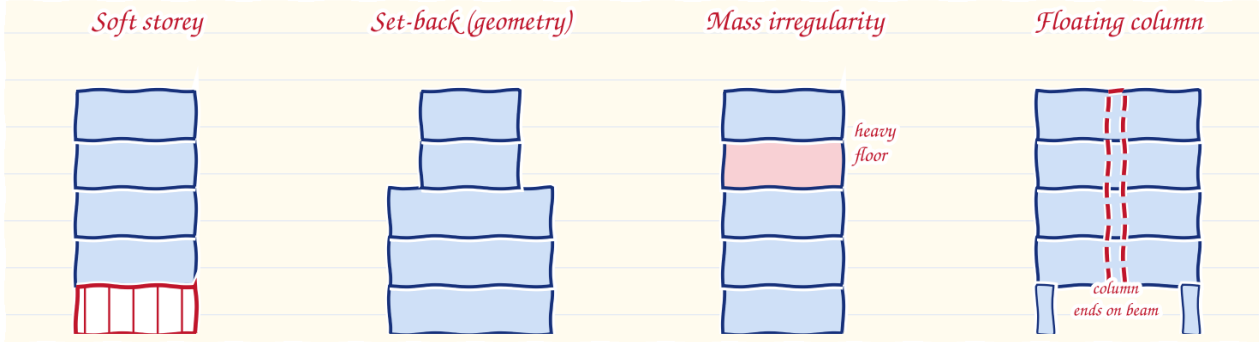


Fig: Vertical irregularities: soft storey, set-back, mass, floating column

1.10 Choice of Construction Materials

Material	Seismic behaviour
Structural steel	Most ductile (yield plateau, elongation 20%+); light weight; high strength/weight -> lower inertia; but local/lateral buckling & connection (weld) failures (Northridge) need care
Reinforced concrete	Good if detailed ductilely (IS 13920); cheap, durable; heavy; cover spall and brittle shear if poorly confined
Masonry	Brittle, low tension/shear strength, heavy; unreinforced masonry is the most vulnerable (needs bands, vertical steel - IS 4326)
Timber	Light, ductile connections; good seismic performance
Earth / adobe	Heavy, weak; need reinforcement or confinement

(a) Unconfined concrete

- Plain concrete is strong in compression but brittle: stress-strain curve rises to f_c at strain ≈ 0.002 and falls quickly; crushing strain ε_{cu} ≈ 0.0035 (IS 456).
- Cover concrete spalls at the end of the strength; sudden loss of capacity.

(b) Confined concrete

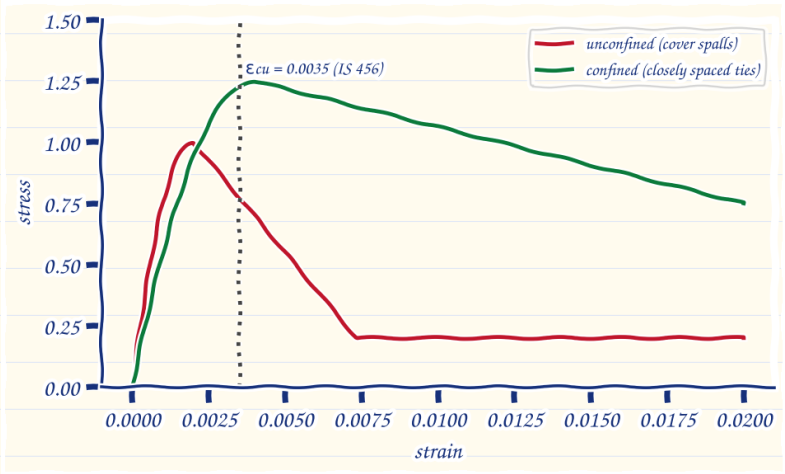


Fig: Confinement increases strength and ductility

- Closely spaced transverse ties/spirals restrain lateral expansion of core -> triaxial compression -> higher strength f_{cc} and much larger ultimate strain (0.01-0.025) - slower descending branch.
- Effectiveness increases with smaller spacing, larger volumetric ratio ρ_s, 135° hooks, circular hoops (best), and supplementary cross-ties.
- Confinement also prevents buckling of longitudinal bars and gives shear strength.

(c) Masonry and (d) Reinforcing steel

- Masonry: good in compression only; tension strength ≈ 0; shear strength depends on mortar and vertical load. Use richer mortar (1:4 or 1:6 cement-sand), strong units (brick ≥ 3.5 MPa), reinforcement & bands.
- Reinforcing steel (IS 13920:2016): Fe 415, Fe 500, Fe 550 (and D grades with better ductility); elongation ≥ 14.5%, actual yield strength not more than 20 N/mm² above nominal, UTS/YS ≥ 1.15 -> ensures ductile yielding and strain hardening. Prefer TMT bars; avoid cold-twisted (CTD) bars.

- Minimum concrete grade: M20 (M25 for buildings > 15 m height in Zones III, IV, V under IS 13920:2016). Avoid very high-strength concrete for ductility reasons (more brittle).

PART B : INTRODUCTION TO EARTHQUAKE RESISTANT D

2.1 Seismic Design Requirements

- Strength: members and connections to resist design forces (limit state design with factored loads, IS 456).
- Stiffness: storey drift ≤ 0.004 h (IS 1893:2016) under design lateral force (DBE, partial load factors 1.0).
- Ductility: IS 13920 detailing in Zones III-V (and Zone II for important buildings).
- Stability: overturning, sliding, P-delta, foundation (liquefaction, differential settlement).
- Integrity: all parts tied together; non-structural elements anchored.
- Both horizontal & vertical (for long spans, cantilevers, prestressed members) shaking considered; torsion included.

2.2 Regular and Irregular Configurations (IS 1893 (Part 1):2016, Cl. 7.1)

Regular building = simple, symmetric, uniform in plan and elevation. Irregular = having any of the following irregularities; it must be analysed dynamically if taller than limits, and R may be reduced / extra checks applied.

Plan irregularity (Table 5)	Limit
Torsional irregularity	Max drift at one end > 1.5 x average / min drift of the two ends
Re-entrant corners	Projection beyond re-entrant corner > 20% of plan dimension in that direction
Diaphragm discontinuity	Openings/cut-outs > 50% of gross floor area, or sudden change in stiffness > 50%
Out-of-plane offsets	Discontinuities in lateral force path (offset of vertical elements)
Non-parallel systems	Vertical lateral-force elements not parallel/symmetric to major orthogonal axes

Vertical irregularity (Table 6)	Limit
Stiffness irregularity - soft storey	Lateral stiffness < 70% of storey above or < 80% of average of three storeys above
Mass irregularity	Seismic weight of any storey > 150% of adjacent storey (roof excepted)
Vertical geometric irregularity	Horizontal dimension of lateral system > 125% of adjacent storey
In-plane discontinuity	Offset of lateral-force resisting element > its length
Strength irregularity - weak storey	Storey lateral strength < 80% of the storey above
Floating / stub columns	Columns not carried to foundation; very short columns

2.3 Basic Assumptions (IS 1893)

- Earthquake causes impulsive ground motions, which are complex and irregular; they do not occur simultaneously with maximum wind or maximum flood or maximum sea waves/current.
- Design EQ force is not combined with wind; loads: EQ + dead + part live.
- Earthquake forces are idealised as equivalent static lateral forces at floor levels (for regular, low-mid rise) - dynamic analysis otherwise.
- Modulus of elasticity of materials, wherever required, may be taken as for static analysis, unless a dynamic value is available (RC: $E_c = 5000\sqrt{f_{ck}}$). Floors/diaphragms rigid in plane.
- Soil-structure interaction usually ignored except for special cases; seismic weight W = DL + fraction of IL.

2.4 Design Earthquake Loads - Seismic Weight

Imposed (live) load on floor	Fraction of IL included in seismic weight
Up to 3.0 kN/m²	25%
Above 3.0 kN/m²	50%
Roof	0% (live load on roof not considered)

Design horizontal seismic coefficient (IS 1893:2016 cl. 6.4.2)

$A_h = (Z/2) \times (S_a/g) / (R/I)$

Z = zone factor (0.10, 0.16, 0.24, 0.36); I = importance factor; R = response reduction factor; S_a/g = design spectrum (5% damping)

Design base shear $V_B = A_h W$ (W = seismic weight of building)

Factor	Values
Importance factor I	1.5 - important service & community buildings (hospital, fire station, power, telecom); 1.2 - business & residential > 200 persons (assembly halls, malls); 1.0 - all other
Response reduction R	OMRF 3.0; SMRF 5.0; ordinary RC shear wall 3.0; ductile shear wall 4.0; shear wall + SMRF dual 4.5-5.0; unreinforced masonry 1.5; reinforced masonry 3.0 (as per IS 1893 Table 9)
Soil types (Sa/g)	Type I rock/hard; Type II medium; Type III soft - different plateaus & decay (see spectrum)
Vertical seismic	Vertical acceleration = 2/3 of horizontal design spectrum; needed for cantilevers > 1.5 m, prestressed beams, long spans > 20 m (IS 1893:2016)

Important / Exam point

Time period (empirical, IS 1893:2016 cl 7.6.2): RC moment frame without infill: $T_a = 0.075\ h^{0.75}$; steel: $T_a = 0.085\ h^{0.75}$; RC/steel frame with brick infill (or shear wall building): $T_a = 0.09\ h/\sqrt{d}$ (h = height in m, d = base dimension in m along the considered direction).

2.5 Basic Load Combinations (IS 1893:2016 / IS 456 limit state)

Limit state of collapse (IS 456 / 1893)	Combination
Gravity only	1.5 (DL + IL)

Limit state of collapse (IS 456 / 1893)	Combination
With EQ	1.2 (DL + IL ± EL)
	1.5 (DL ± EL)
Uplift / overturning check	0.9 DL ± 1.5 EL
Earthquake in both directions	EL = ± EL_x ± 0.3 EL_y or ± 0.3 EL_x ± EL_y (+ vertical where needed)

- Working stress method: DL + IL ± EL, DL ± EL with allowable stresses increased by 33.3%.
- In IS 1893 EL includes combination of all three components if required: ±EL_x ± 0.3EL_y ± 0.3EL_z.

2.6 Permissible Stresses

- When WSM is used, permissible stresses in materials (concrete, steel, masonry) and allowable soil bearing pressure may be increased by 33.33% for EQ load combinations (IS 1893/IS 456 Annex B).
- For limit state design, partial safety factors: γ_c = 1.5 (concrete), γ_s = 1.15 (steel) generally, but for seismic combos γ_c = 1.5 and γ_s = 1.15 are retained while load factor is reduced (1.2 / 1.5).
- Soil: for liquefiable soils increase not allowed; for soft soils (N < 10) - see Table of IS 1893 for allowable bearing capacity increase.

2.7 Seismic Methods of Analysis

Method	Basis	When used
Equivalent static (ELF / seismic coefficient)	Single mode; $V_B = A_h W$ distributed with height	Regular buildings: Zone II/III up to 90 m; Zone IV/V up to 40 m (regular); irregular buildings < 12 m (IV, V) / < 40 m (II, III)
Response spectrum method	Modal analysis, modes combined by SRSS/CQC; ≥ 90% mass participation	Regular > limits, all irregular buildings above limits
Time history analysis	Step-by-step integration with accelerograms (linear or non-linear)	Important / very tall / irregular buildings, base-isolated, non-linear studies
Pushover (static non-linear)	Incremental lateral load until collapse mechanism	Performance-based design, retrofit evaluation

Design base shear from dynamic analysis V_B (dynamic) must be scaled up to the base shear V_B from empirical period of ELF (so that $V_B \geq V_B$).

2.8 Factors in Seismic Analysis

- Z - seismic zone; I - importance; R - ductility/overstrength reduction; S_a/g - soil & period dependent; W - seismic weight; h, d - height and base dimension for T.
- Soil type and foundation (liquefaction, bearing).
- Damping (5% standard; other values via multiplying factor); Torsion (accidental eccentricity 0.05 b); P-Δ effects; Orthogonal effects; Drift limits.
- Redundancy and over-strength: well-distributed lateral system is better.

2.9 Equivalent Lateral Force Method - Procedure

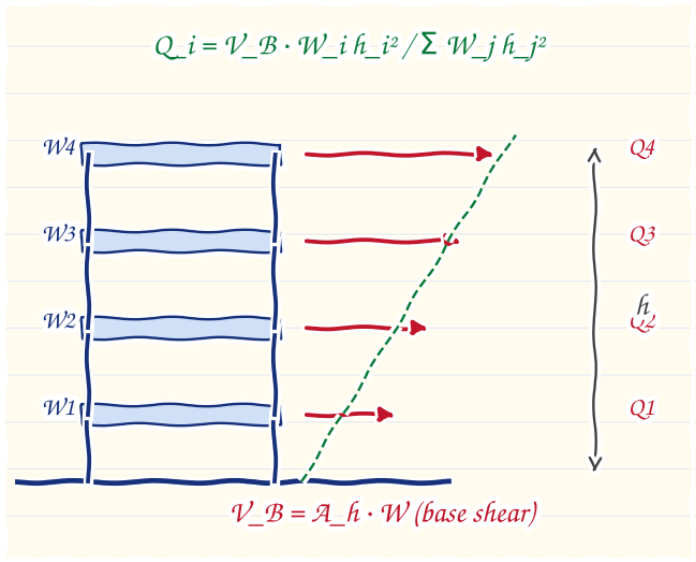


Fig: Equivalent lateral force distribution

1. Find seismic weights W_i of each floor (DL + 25/50% IL; roof IL = 0) and total W.
2. Find fundamental natural period T_a in each direction (empirical formulas above).
3. Read S_a/g for T_a and soil type from the design spectrum (e.g., medium soil, $T > 0.55$: $S_a/g = 1.36/T$, capped 2.5).
4. Compute $A_h = (Z/2)(S_a/g)/(R/I)$ and $V_B = A_h W$.
5. Distribute along height: $Q_i = V_B W_i h_i^2 / \Sigma W_j h_j^2$ (parabolic distribution).
6. Distribute to lateral-load resisting elements (frames/walls) by stiffness; add torsion; check drift ≤ 0.004 h, P-delta, overturning.
7. Analyse for load combinations, design members, then detail as per IS 13920.

Solved example (ELF)

4-storey RC office building, Zone III ($Z = 0.16$), medium soil, SMRF ($R = 5$), $I = 1.0$; $W = 4 \times 3000$ kN = 12 000 kN; $h = 13$ m; $d = 20$ m; infilled frame.

$T_a = 0.09\, h/\sqrt{d} = 0.09 \times 13/\sqrt{20} = 0.262$ s (< 0.55 s plateau) -> $S_a/g = 2.5$

$A_h = (0.16/2)(2.5)/(5/1) = 0.08 \times 2.5/5 = 0.04$

$V_B = 0.04 \times 12\,000 = 480$ kN (4% of weight)

Continued

Distribution of Q_i (equal floors of 3 m): Q_i proportional to $h_i^2 = 3^2, 6^2, 9^2, 12^2 = 9, 36, 81, 144$ (sum 270): $Q = 480 \times (9, 36, 81, 144)/270 = 16, 64, 144, 256$ kN (check: sum 480 kN).

EXTRA POINTS & NUMERICAL

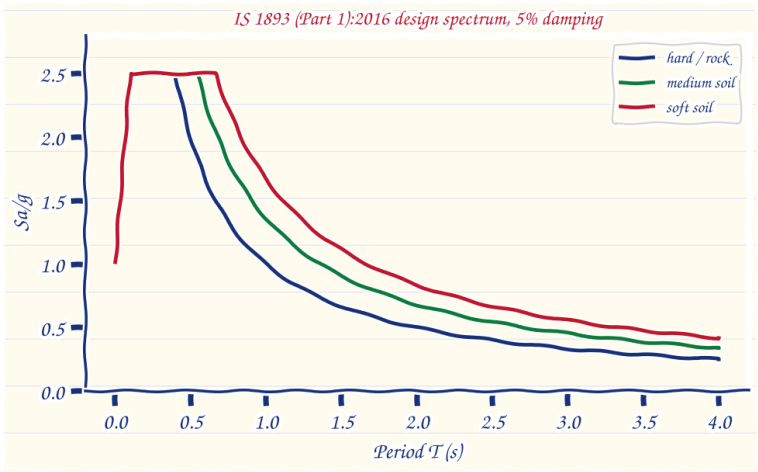


Fig: Design spectrum S_a/g (5% damping) - IS 1893 (Part 1):2016

Soil type	$0 \le T \le 0.10$	Plateau	Decay (up to $T = 4$ s)	$T > 4$ s
Type I - Rock / hard soil	$1 + 15T$	2.5 (0.10-0.40 s)	$1.00/T$	0.25
Type II - Medium soil	$1 + 15T$	2.5 (0.10-0.55 s)	$1.36/T$	0.34
Type III - Soft soil	$1 + 15T$	2.5 (0.10-0.67 s)	$1.67/T$	0.42

Common seismic problems	Why it happens	Remedy
Soft storey	Open ground floor (parking) has less stiffness; all drift concentrates in one storey	Provide RC shear walls or bracing in the open storey; avoid it altogether where possible
Short column	Partial-height infill or mezzanine -> column stiffer than its neighbours attracts more shear -> shear failure	Full-height gap between infill and column, or ductile shear detailing over full height
Pounding	Adjacent buildings of different periods hit each other	Provide seismic gap ($R \times \Delta$)
Torsion	CM ≠ CR	Symmetric layout, peripheral walls
Weak beam-column joint	Shear/anchorage failure at joint	Transverse ties continue through joint (IS 13920)
Heavy parapets/tanks	Large mass at top, whiplash	Reduce mass, anchor, brace

Solved example 2 (hospital, Zone V)

Zone V ($Z = 0.36$), hospital ($I = 1.5$), SMRF ($R = 5$), rock site, $T_a = 0.40$ s, $W = 20\,000$ kN.

$S_d/g = 2.5$ (plateau); $A_h = (0.36/2)(2.5)/(5/1.5) = 0.45/3.333 = 0.135$

$V_B = 0.135 \times 20\,000 = 2700$ kN (13.5% of weight) - for an ordinary building ($I = 1$) it would have been 1800 kN.

Important / Exam point

Difference OMRF vs SMRF: OMRF ($R = 3$) is not detailed per IS 13920 -> allowed only in Zone II; SMRF ($R = 5$) has ductile detailing and is mandatory in Zones III, IV, V. Higher R gives lower design force but demands better detailing.

QUICK REVISION - Likely questions

1. Explain the seismic design philosophy and the four virtues of an earthquake-resistant building.
2. Importance of symmetry, simplicity; effect of plan/elevation irregularities - with sketches.
3. What is ductility? Explain ductility relationships and how it is achieved; confined vs unconfined concrete.
4. Explain torsion in buildings and ways to reduce it; soft-storey and short-column effect.
5. Classify irregularities as per IS 1893 (Part 1):2016; limits.
6. Basic assumptions and load combinations for seismic design; permissible stresses.
7. Explain methods of seismic analysis and conditions of applicability of ELF.
8. Procedure of ELF method; numerical for base shear and its distribution.

UNIT - III

Reinforced Concrete Buildings

Syllabus (as given):

Principles of earthquake resistant design of RC members - structural models for frame buildings - seismic methods of analysis - IS code based methods - vertical irregularities - plan configuration problems - lateral load resisting systems - design lateral forces as per IS 1893 (Part 1):2016 - equivalent lateral force procedure - lateral distribution of base shear.

1. PRINCIPLES OF EQ RESISTANT DESIGN OF RC MEMBE

RC buildings are the most common structures in India. Their performance in an earthquake depends on how well the members are proportioned and detailed to be ductile. Design = (i) get forces from analysis (IS 1893), (ii) design sections (IS 456), (iii) detail for ductility (IS 13920:2016).

- *Strong column - weak beam: sum of column flexural capacities at a joint ≥ 1.4 x sum of beam capacities (IS 13920:2016) so plastic hinges form in beams.*
- *Capacity design for shear: design shear force V_e = shear from gravity + shear from plastic moment capacities at ends ($V_e = V_g \pm 1.4\,(M_u,sag + M_u,hog)/L$) - ensures flexural (ductile) failure occurs before shear (brittle).*
- *Confinement of concrete at member ends (plastic hinge zones) with closely spaced hoops with 135° hooks.*
- *Adequate anchorage and lap splices located away from high-moment zones (columns: central half); hooks inside the confined core.*
- *Joint integrity: beam-column joints must stay elastic; transverse reinforcement continued through joint; beam bars properly anchored.*
- *Under-reinforced sections: tension steel yields before concrete crushes (limit ρ_{max} = 2.5% in beams; columns 0.8%-4%).*
- *Keep structure redundant: many bays of moment frames so that failure of one hinge does not collapse the structure.*

Member	Seismic role	Key IS 13920:2016 provisions
Beam	Dissipates energy by flexural hinging	$b \geq 200$ mm, $b/D \geq 0.3$, $D \leq L/4$; $\rho_{min} = 0.24\sqrt{f_{ck}}/f_y$, ρ_{max} = 2.5%; min 2 bars top & bottom throughout; stirrups with 135° hook, spacing ≤ d/4, 8φ, 100 mm in 2d zone
Column	Must remain elastic (axial + bending)	Min size 300 mm, $b/D \geq 0.4$; 0.8% ≤ ρ ≤ 4%; closely spaced hoops over l_0 at ends; $\Sigma M_c \geq 1.4\,\Sigma M_b$
Joint	Transfers forces	Hoops continued through joint; special confining steel; anchorage beyond joint core
Slab (diaphragm)	Distributes inertia forces	Monolithic with beams; openings < 50% area; adequate chords/collectors

Member	Seismic role	Key IS 13920:2016 provisions
Shear wall	Main lateral load resisting	Min thickness 150 mm; ρ_{min} 0.25% each way; boundary elements if needed

2. STRUCTURAL MODELS FOR FRAME BUILDINGS

- 3-D frame model (space frame) - most accurate; beam and column elements (with rigid end zones), slab as rigid/semi-rigid diaphragm; software like ETABS/STAAD.
- Plane frame model - for regular, symmetric buildings; each frame analysed separately for its share of lateral load.
- Lumped mass shear building model - floors rigid, one lateral DOF per floor (per direction) - basis for modal analysis. Stiffness of each storey $k_i = \Sigma 12EI/h^3$ (columns fixed-fixed).
- Equivalent frame / equivalent strut for infill: masonry infill replaced with a diagonal compression strut (width $w_{ds} = 0.175\,\alpha_h^{-0.4}\,L_{ds}$, IS 1893:2016 cl 7.9.2.2).
- Modelling points: cracked section properties (beams 0.35 I_g, columns 0.7 I_g - IS 1893:2016 cl 6.4.3.1 for deflection/period); rigid joints; fixed vs pinned base; P-Δ; soil flexibility; seismic weight lumped at floor levels (with 25-50% IL).
- Number of modes in dynamic analysis: such that total modal mass participation ≥ 90% in each direction.

> Cracked stiffness
> IS 1893:2016: for RC members effective stiffness (cracked): beams 0.35 EI_g, columns 0.70 EI_g, walls 0.70 EI_g (uncracked), 0.35 (cracked) - for analysis under seismic loading.

3. SEISMIC METHODS OF ANALYSIS (IS CODE BASED)

Method	Applicability (IS 1893:2016 cl. 7.6)
Equivalent static (seismic coefficient)	Regular buildings: Zone II & III up to 90 m, Zone IV & V up to 40 m; irregular buildings up to 12 m (IV, V) or 40 m (II, III)
Dynamic analysis (response spectrum / time history)	All others - tall regular, irregular above limits; mandatory for buildings with vertical irregularity, high torsion, special structures

- Response spectrum method: (1) find mode shapes & periods (eigen analysis), (2) for each mode obtain S_a/g, (3) modal base shear $V_k = A_k W_k$, (4) combine modes with SRSS (well-separated modes) or CQC (closely spaced modes, frequency difference < 10%).
- Time-history: accelerograms scaled to the design spectrum; direct integration (Newmark β) of equations of motion - for important structures.
- Dynamic base shear V_B must be ≥ base shear from empirical period; otherwise scale up all results by V_B/V_B.
- Orthogonal effects: 100% in one direction + 30% in the perpendicular direction (and vice versa), take the worst.

4. VERTICAL IRREGULARITIES

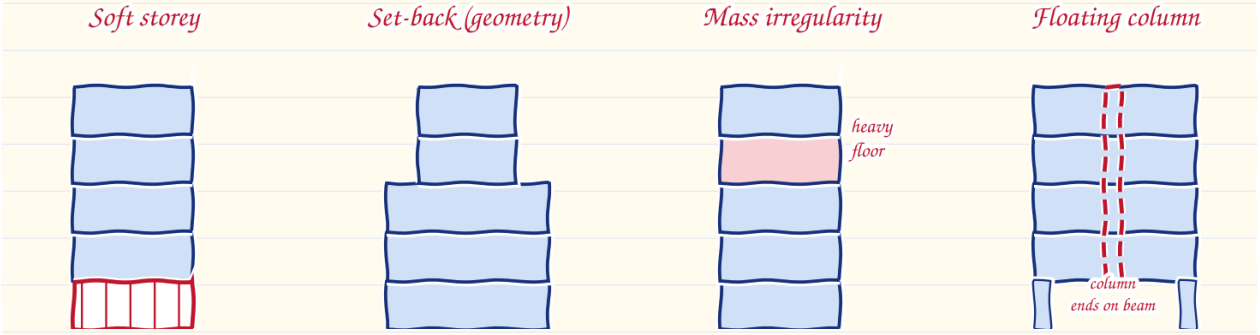


Fig: Common vertical irregularities

- Soft storey (stiffness irregularity) - ground-floor parking, tall ground storey, no infill; drift concentrates -> pancake collapse (Bhuj 2001).
- Weak storey (strength < 80% of above). Mass irregularity (> 150% of neighbouring floors). Vertical geometric irregularity / set-back (> 125%).
- In-plane discontinuity (shear wall not reaching foundation), floating columns (transfer girders, discontinuous load path).
- Short column effect: columns restrained by partial-height infills/mezzanines/split-level floors have a short free length -> very stiff -> attract large shear -> X-shaped shear cracks.
- Consequences: concentration of damage, inelastic demand over limited elements, sudden failure. IS 1893:2016: such buildings need dynamic analysis; soft storey shall be avoided (RC walls added in ground storey).

5. PLAN CONFIGURATION PROBLEMS

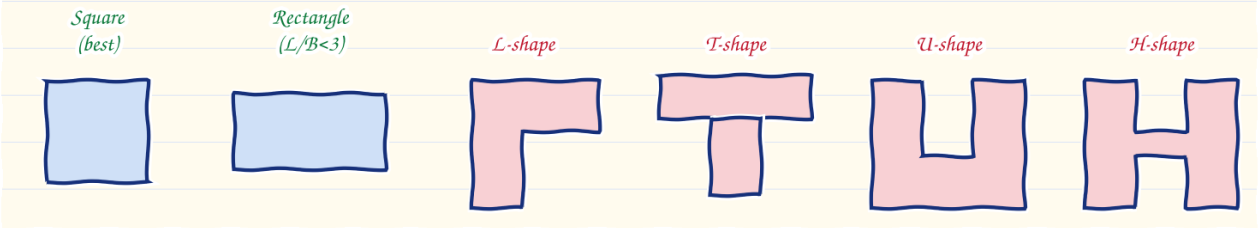


Fig: Plan shapes: good vs poor

- Torsion (CM ≠ CR): corner/edge columns heavily loaded.
- Re-entrant corners (L, T, U, H, + shapes): stress concentration at the inner corner; wings vibrate differently -> use seismic joints.
- Diaphragm discontinuity: large openings (staircase, atrium, lifts) weaken floor action; keep openings < 50% and away from corners/walls, provide trimmer steel.
- Out-of-plane offsets and non-parallel systems: eccentric load paths -> torsion.
- Pounding between adjacent buildings - provide gap; expansion/separation joints in long buildings (> 45-50 m).
- Remedies: symmetric simple shape; distribute walls symmetrically at periphery; limit L/B ≤ 3; separate wings with joints; avoid heavy masses far from CR.

6. LATERAL LOAD RESISTING SYSTEMS

System	How it works	R (IS 1893:2016)	Notes
Ordinary RC moment frame (OMRF)	Frame action, no special detailing	3.0	Zone II only
Special RC moment frame (SMRF)	Ductile frame (IS 13920)	5.0	Zones III-V; up to ~10-12 storeys economical
Ordinary RC shear wall (with OMRF)	Walls resist lateral load	3.0 / 3.5	Stiff; limited ductility
Ductile RC shear wall	Walls detailed per IS 13920	4.0	Efficient 10-40 storeys
Dual (wall + SMRF)	Frames ≥ 25% of base shear designed to share	4.5-5.0	Redundant; best
Steel braced frames, EBF	Axial/shear link yielding	4.0 - 5.0	Steel buildings
Unreinforced masonry (load bearing)	Walls	1.5	Low-rise only (see Unit IV)

- *Moment frames*: flexible, architecturally open; drift governs design; strong-column weak-beam.
- *Shear walls*: resist most lateral load by flexure + shear; reduce drift; need boundary elements; place symmetrically.
- *Dual systems*: walls take the majority in lower storeys, frames take it in the upper storeys; interaction gives nearly uniform drift.
- Floor diaphragms (RC slabs) tie everything together and distribute inertia forces by stiffness.

7. DETERMINATION OF DESIGN LATERAL FORCES - IS 18

Step A. Seismic weight

$W = \Sigma W_i$ with W_i = DL (slab, beams, columns half above/below, walls, finishes) + 25% IL (IL ≤ 3 kN/m²) or 50% IL (IL > 3 kN/m²); roof live load = 0.

Weight of columns/walls in a storey: half above + half below the floor.

Step B. Natural period (empirical, cl. 7.6.2)

RC moment frame (bare, no infill): $T_a = 0.075\ h^{0.75}$

Steel moment frame: $T_a = 0.085\ h^{0.75}$

RC/steel frame with brick infill, shear wall buildings, others: $T_a = 0.09\ h/\sqrt{d}$

h = building height (m) from foundation/ground (excluding basement if walls rigid); d = base dimension (m) in the direction of EQ.

Step C. Design horizontal seismic coefficient

$A_h = (Z/2)(S_a/g) / (R/I)$

S_a/g from the design spectrum for the soil type (I/II/III) at T_a, 5% damping; use $S_a/g = 2.5$ on the plateau.

Code lower bound: $A_h \geq Z/2 \times 1/(R/I) \times 0.25$-type minimum is implicit in $S_a/g = 0.25$ for $T > 4$ s.

Step D. Design base shear

$V_B = A_h W$

8. EQUIVALENT LATERAL FORCE PROCEDURE & LATER

$$Q_i = V_B \cdot W_i h_i^2 / \Sigma W_j h_j^2$$

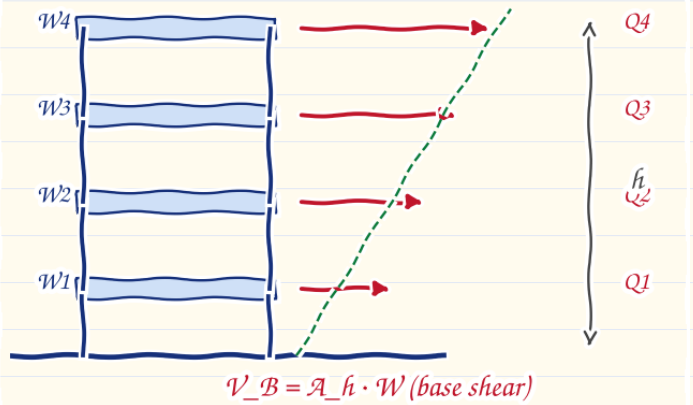


Fig: Lateral distribution of base shear Q_i along height

Vertical distribution (cl. 7.6.3)

$Q_i = V_B \times W_i h_i^2 / \Sigma_{j=1}^{n} W_j h_j^2$

Q_i = design lateral force at floor i; W_i = seismic weight of floor i; h_i = height of floor i measured from base (m).

$\Sigma Q_i = V_B$. Storey shear $V_i = \Sigma$ of Q_j above storey i; overturning moment $M_i = \Sigma Q_j (h_j - h_i)$.

- Lateral force distribution is parabolic (quadratic in height) - assumes fundamental mode shape is linear.
- Horizontal distribution at each floor to frames/walls: (a) rigid diaphragm -> in proportion to relative lateral stiffness (rigidity) of the vertical elements; (b) flexible diaphragm -> by tributary area/mass.
- Torsion: add torsional moment $M_t = Q_i \times e_{di}$ with design eccentricity $e_{di} = 1.5\, e_{si} + 0.05\, b_i$ or $e_{si} - 0.05\, b_i$ (whichever is more severe).
- Storey drift limit: ≤ 0.004 h under design lateral force (with partial factor 1.0), computed with cracked sections; no need of P-Δ if stability index $\theta = P\Delta/(V h) < 0.1$.

- Frame lateral stiffness of one column: $k = 12EI/h^3$ (both ends fixed); sum over columns of storey -> storey stiffness; portal/cantilever methods for approximate analysis of frames.

Worked Example (full procedure)

Problem

G+3 RC school building (Zone IV, $Z = 0.24$; $I = 1.5$ [important]; SMRF $R = 5$; medium soil). Plan 15 m x 12 m, storey height 3.2 m, height $h = 12.8$ m (4 floors), brick infill.

Seismic weights: floor 1,2,3 = 3200 kN each, roof = 2400 kN -> $W = 12\,000$ kN.

Direction of shorter dimension $d = 12$ m: $T_a = 0.09 \times 12.8/\sqrt{12} = 0.332$ s (< 0.55 s) -> $S_d/g = 2.5$

$A_h = (0.24/2) \times 2.5/(5/1.5) = 0.12 \times 2.5/3.333 = 0.09$

$V_B = 0.09 \times 12\,000 = 1080$ kN

Floor	W_i (kN)	h_i (m)	$W_i h_i^2$	$W_i h_i^2/\Sigma$	Q_i (kN)	Storey shear (kN)
Roof (4)	2400	12.8	393 216	0.4615	498.5	498.5
3	3200	9.6	294 912	0.3462	373.9	872.4
2	3200	6.4	131 072	0.1538	166.2	1038.6
1	3200	3.2	32 768	0.0385	41.5	1080.1
Σ	12 000		851 968	1.000	1080	

Check

Overturning moment at base = $\Sigma Q_i h_i = 498.5 \times 12.8 + 373.9 \times 9.6 + 166.2 \times 6.4 + 41.5 \times 3.2 = 6381 + 3589 + 1064 + 133 = 11\,167$ kN.m. Each floor force is then shared among frames in proportion to their stiffness, plus torsion.

9. MORE THEORY YOU NEED FOR THE EXAM

9.1 Response reduction factor R

Actual structures have reserve strength and ductility, so they can be designed for a force much lower than the elastic force. $R = R_\mu \times \Omega \times R_r$ (ductility x over-strength x redundancy).

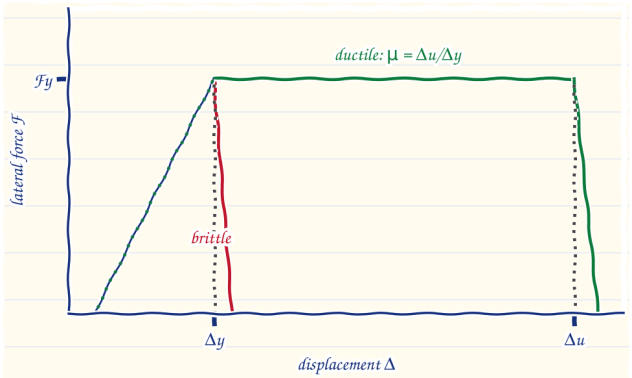


Fig: Elastic force vs design force: reduction by ductility

- Elastic base shear (if structure stayed elastic) $V_{el} = R \times V_{design}$ (e.g., $R = 5$ -> $V_{el} = 5\,V_B$).

- R_μ (ductility reduction): equal-displacement $R_\mu = \mu$ ($T > 0.5$ s); equal-energy $R_\mu = \sqrt{(2\mu - 1)}$ (short period).
- Ω (over-strength) 1.5 - 2.0: actual strength > design strength due to material strength, load factors, minimum steel, slab participation.
- R_r (redundancy): many hinges -> load shared; low redundancy buildings get smaller R.
- Hence IS 1893 gives smaller R to brittle systems (URM 1.5) and larger R to ductile systems (SMRF 5).

9.2 Seismic weight - how to compute (floor example)

Floor 15 m × 12 m (180 m²), live load 3 kN/m², slab 150 mm

Slab 0.15 × 25 = 3.75 kN/m²; floor finish 1.0; partitions 1.0 -> DL = 5.75 kN/m² -> 5.75 × 180 = 1035 kN

Beams (say 0.3 × 0.5 m, total 130 m): 0.3 × 0.35 (below slab) × 25 × 130 = 341 kN

Columns: 20 nos 0.4 × 0.4, storey 3.2 m, half above & below = 3.2 m: 20 × 0.16 × 3.2 × 25 = 256 kN

Outer walls (230 mm, 3.0 m, 54 m perimeter, 19 kN/m³): 0.23 × 3 × 54 × 19 = 708 kN

Live load share: 25% × 3 × 180 = 135 kN (IL ≤ 3 kN/m²)

W_{floor} = 1035 + 341 + 256 + 708 + 135 = 2475 kN (roof: no IL; parapet adds weight).

9.3 Lateral stiffness of columns/frames and distribution to frames

Column stiffness (fixed-fixed, cracked 0.7 EI_g)

$E_c = 5000\sqrt{f_{ck}} = 5000\sqrt{25} = 25\,000$ N/mm² = 25×10^6 kN/m²; column 400 × 400: $I_g = 0.4 \times 0.4^3/12 = 2.133 \times 10^{-3}$ m⁴

$k = 12\,(0.7\,E\,I_g)/h^3 = 12 \times 25e6 \times 1.493e\text{-}3/3.0^3 =$ 16 590 kN/m per column (h = 3 m)

Storey stiffness $K = \Sigma\, k$ of all columns in that storey (parallel springs).

Distribution of floor force to frames (rigid diaphragm, no torsion)

Floor force Q = 360 kN; Frame A, B, C stiffness: 20 000, 30 000, 20 000 kN/m (Σ = 70 000)

$Q_A = 360 \times 20/70 =$ 102.9 kN; $Q_B = 360 \times 30/70 =$ 154.3 kN; $Q_C =$ 102.9 kN -> shares ∝ relative rigidity

Remember

A stiffer element attracts more load. This is why a stiff short column fails first (short-column effect) and why walls take most load in a dual system.

9.4 Torsion - numerical

Eccentricity calculation

Plan 20 m (x) x 10 m; EQ force acts in y-direction, b = 20 m; CM at x = 10.0 m, CR at x = 11.2 m -> e_s = 1.2 m

e_{d1} = 1.5 e_s + 0.05 b = 1.8 + 1.0 = 2.8 m; e_{d2} = e_s - 0.05 b = 1.2 - 1.0 = 0.2 m -> use the worse effect for each element

Torsional moment at the floor M_t = Q_i x e_d = 400 x 2.8 = 1120 kN.m; extra shear in a frame at distance x from CR = M_t k_x x / J, J = Σ $k x^2$ (torsional rigidity).

9.5 Beam-column joint checks (capacity design)

Strong column - weak beam (IS 13920:2016: $\Sigma M_c \geq 1.4 \Sigma M_b$)

At a joint: beam hogging capacity 210 kN.m (left) and sagging 130 kN.m (right): ΣM_b = 340 kN.m

Columns above and below: 280 kN.m each -> ΣM_c = 560 kN.m; required 1.4 x 340 = 476 kN.m -> 560 ≥ 476 OK

Design shear force for beam (capacity design)

Clear span L = 6 m; V_{ug} (gravity, factored 1.2(DL+IL)) = 60 kN; M_u,lim sag = 130 kN.m; M_u,lim hog = 210 kN.m

V_e = V_{ug} + 1.4 (M^{sag} + M^{hog})/L = 60 + 1.4 (130 + 210)/6 = 60 + 79.3 = 139.3 kN (design stirrups for this V_e, not the analysis shear)

9.6 Mode combination in response spectrum method

- *Modal participation factor P_k = Σ $W_i \varphi_{ik}$ / Σ $W_i \varphi_{ik}^2$; modal mass M_k = (Σ $W_i \varphi_{ik}$)²/(g Σ $W_i \varphi_{ik}^2$).*
- *Modal base shear V_k = (Z/2)(S_a/g)$_k$/(R/I) x M_k g; lateral force in mode k at floor i: Q_{ik} = $A_k \varphi_{ik} P_k W_i$.*
- *SRSS: V = √(V_1^2 + V_2^2 + V_3^2 + ...); e.g., V_1 = 800, V_2 = 250, V_3 = 90 kN -> V = √(640 000 + 62 500 + 8 100) = 843 kN*
- *CQC when modal frequencies differ by < 10% (3-D, torsionally coupled buildings): V = √(ΣΣ $\rho_{ij} V_i V_j$).*
- *Include enough modes to capture ≥ 90% of seismic mass in each orthogonal direction.*

9.7 Drift, P-Δ and stability

- *Storey drift Δ_i = u_i - u_{i-1} ≤ 0.004 h_i (e.g., 12.8 mm for h = 3.2 m) under the design lateral force with load factor 1.0 and cracked sections (IS 1893:2016 cl. 7.11.1.1).*
- *P-Δ (second-order) effect: gravity load acting through lateral displacement adds moment; stability index θ = P_u Δ/(V_u h); if θ ≤ 0.1 neglect, θ > 0.1 amplify forces by 1/(1 - θ); θ < 0.33 limit.*
- *Minimum design lateral force: $V_B \geq V_B$; torsion and orthogonal effects included.*

9.8 Dual system - 25% rule and interaction

- In a dual system the moment frames (SMRF) must be designed to resist at least 25% of the design base shear independently, while walls + frames together resist 100%.
- Wall behaves like a vertical cantilever (flexural shape), frame like a shear beam (shear shape): they push and pull each other - walls carry load in lower storeys, frames in upper storeys, giving a more uniform drift profile.
- Wall-frame stiffness ratio influences the distribution; tie them with rigid diaphragms and coupling beams.

9.9 Soft-storey check example

Storey	Lateral stiffness (kN/mm)	% of above	% of avg of 3 above	Result
4	95	-	-	-
3	100	105%	-	OK
2	98	98%	-	OK
1 (ground)	52	53% (< 70%)	52/97.7 = 53% (< 80%)	Soft storey!

Ground storey stiffness 52 kN/mm is < 70% of storey above and < 80% of the average of three above (97.7) -> vertical stiffness irregularity: provide RC shear walls/bracing at ground floor or revise the frame.

9.10 Design process flow (RC building to IS 1893/13920)

1. Fix configuration (regular, symmetric), decide system (SMRF / dual), zone & soil.
2. Preliminary member sizes (beam $b \geq 200$, $D \approx L/12$; column ≥ 300; check against IS 13920 limits).
3. Compute seismic weights; empirical T_a; S_d/g; A_h; V_B; distribute Q_i; torsion.
4. 3-D analysis (ETABS/STAAD) for DL, IL, EL_x, EL_y; check mass participation, drift 0.004 h, torsion irregularity.
5. Combine loads (1.2/1.5/0.9); design beams (flexure + capacity shear), columns (P-M, biaxial), joints, slabs, foundations.
6. Detail as per IS 13920:2016 (hoops, anchorages, laps) - see Unit V.
7. Check non-structural elements, seismic joints, pounding, foundation/liquefaction.

9.11 Lessons from past earthquakes - RC buildings

Earthquake	Typical RC failures observed
Bhuj 2001 (India)	Soft ground storeys (parking), pancake collapse of Ahmedabad multi-storeys, short column failures, poor concrete, weak joints
Kobe 1995	Soft/weak storey at mid-level, inadequate stirrups in columns, shear failure
Northridge 1994	Non-ductile frames, steel connection (weld) fractures, parking structure collapse
Mexico City 1985	Resonance of 6-15 storey buildings with soft-clay site period; pounding
Nepal 2015 (Gorkha)	Unreinforced masonry and mid-rise RC with soft storeys, poor detailing

9.12 Worked example - 2-DOF shear building (response spectrum method)

Data

Two floors, $m_1 = m_2 = 100$ t ($W = 981$ kN each), $k_1 = k_2 = 40$ MN/m; $Z/2 = 0.08$, $R/I = 5$, medium soil.

Eigen-problem: $(K - \omega^2 M)\phi = 0 \Rightarrow \omega^2 = (3 \mp \sqrt{5})/2 \times k/m = 0.382$ k/m and 2.618 k/m, k/m = 400 s^{-2}

$\omega_1 = 0.618 \times 20 = 12.36$ rad/s ($T_1 = 0.508$ s); $\omega_2 = 1.618 \times 20 = 32.36$ rad/s ($T_2 = 0.194$ s)

Mode shapes (floor 1, floor 2): $\phi_1 = (1, 1.618)$; $\phi_2 = (1, -0.618)$

Modal participation, mass and base shear

$P_1 = (1 + 1.618)/(1 + 2.618) = 0.724$; $P_2 = (1 - 0.618)/(1 + 0.382) = 0.276$

Modal mass fraction: $M_1 = (2.618)^2/(2 \times 3.618) = 94.7\%$; $M_2 = (0.382)^2/(2 \times 1.382) = 5.3\%$ ($\Sigma = 100\%$)

Both periods are on the plateau (0.10-0.55 s): $S_d/g = 2.5 \Rightarrow A_h = 0.08 \times 2.5/5 = 0.04$ for both modes

$V_1 = 0.04 \times 0.947 \times 1962 = 74.3$ kN; $V_2 = 0.04 \times 0.053 \times 1962 = 4.1$ kN

SRSS: $V_B = \sqrt{(74.3^2 + 4.1^2)} = 74.4$ kN (compare ELF with same A_h: $0.04 \times 1962 = 78.5$ kN - similar since mode 1 dominates).

Feature	Equivalent lateral force (static)	Response spectrum (dynamic)
Modes considered	Fundamental only	All significant modes (≥ 90% mass)
Force distribution	Parabolic $W h^2$ formula	From mode shapes (can have whiplash at top)
Period	Empirical formula	Computed from model (base shear scaled to empirical-period value)
Torsion, irregular buildings	Approximate	Accurate with 3-D model
Effort	Hand calculation	Software needed

9.13 Common mistakes to avoid

- Forgetting that roof live load is not included and that only 25%/50% of floor IL is included in W.
- Using h from the wrong datum, or d in the wrong direction ($T_a = 0.09\, h/\sqrt{d}$ must be done for each direction).
- Using Z instead of $Z/2$ in A_h.
- Not applying I/R correctly: $A_h = (Z/2)(S_d/g)/(R/I) = (Z/2)(S_d/g) \times I/R$.
- Missing accidental eccentricity 0.05 b and orthogonal 30% rule.
- Using gross sections when code requires cracked stiffness for drift and period.

9.14 Foundations and ground effects on RC buildings

- Isolated footings in Zones III-V (IS 4326 / IS 1893): tie them with plinth/tie beams in both directions so that the foundation moves as one unit; avoid mixing foundation types in one

building.

- Liquefaction: saturated loose sand loses strength (pore pressure rises, effective stress -> 0), buildings tilt/sink, lateral spreading; check with SPT N-values; remedies: densification, stone columns, piles through the liquefiable layer, raft.
- Soft soils amplify motion and lengthen the period (Type III spectrum); foundation stiffness and rocking can alter the period (soil-structure interaction).
- Basement walls: stiff box - reduces h used for T_a (take h from the top of the basement slab only if walls are rigid).
- Pile foundations: design for kinematic effects and lateral loads; pile caps tied by beams.

9.15 Formula sheet - Unit III

Quantity	Formula
Design coefficient	$A_h = (Z/2)(S_a/g)/(R/I)$
Base shear	$V_B = A_h W$
Floor force	$Q_i = V_B W_i h_i^2 / \Sigma W_j h_j^2$
Period (RC frame / infill)	$T_a = 0.075\, h^{0.75}$ / $0.09\, h/\sqrt{d}$
Seismic weight	$W = DL + 0.25\ IL\ (IL \le 3)$ / $0.50\ IL\ (IL > 3)$; roof $IL = 0$
Design eccentricity	$e_d = 1.5\, e_s + 0.05\, b$ or $e_s - 0.05\, b$
Drift limit	$\Delta \le 0.004\, h$
Column stiffness	$k = 12EI/h^3$
Capacity shear (beam)	$V_e = V_{ug} \pm 1.4\,(M_{sag} + M_{hog})/L$
Strong column check	$\Sigma M_c \ge 1.4\, \Sigma M_b$
SRSS	$V = \sqrt{(\Sigma V_k^2)}$

9.16 Approximate analysis of a frame - Portal method (example)

Use when a quick hand check of frame forces under Q_i is needed. Assumptions: point of contraflexure at mid-height of columns and mid-span of beams; interior columns carry twice the shear of exterior columns (for a multi-bay frame).

3-storey, 2-bay frame, storey height 3.2 m; floor loads $Q_3 = 60$, $Q_2 = 40$, $Q_1 = 20$ kN

Storey shears: $V_3 = 60$; $V_2 = 100$; $V_1 = 120$ kN

2 bays -> 3 columns: shear per exterior column = $V/4$, interior = $V/2$: storey 3: 15 / 30 / 15; storey 2: 25 / 50 / 25; storey 1: 30 / 60 / 30 kN

Column end moments = $V_{col} \times h/2 = V_{col} \times 1.6$: storey 3: 24 / 48 / 24; storey 1: 48 / 96 / 48 kN.m

Beam moment at an exterior joint = column moments meeting there (top: $M_{beam} = M_{col}$ top end); at interior joints the sum of column moments is shared equally by the two beams.

Beam shear $V_b = (M_{left} + M_{right})/L_b$; column axial forces from overturning (sum of beam shears).

> *Note*
> *Portal method is accepted only for preliminary design/checks. The code requires a full 3-D analysis, but the idea (storey shear divided among columns by stiffness) is the same.*

QUICK REVISION - Likely questions

1. *State principles of earthquake resistant design of RC members; explain strong column - weak beam and capacity design.*
2. *Describe structural models for frame buildings (shear building, 3-D frame, infill strut).*
3. *Explain IS code based methods of analysis and conditions for dynamic analysis.*
4. *Explain vertical irregularities and plan configuration problems with sketches and remedies.*
5. *Compare lateral load resisting systems: moment frame, shear wall, dual system; R values.*
6. *Calculate design base shear as per IS 1893 (Part 1):2016 and distribute it along the height (numerical).*

UNIT - IV

Masonry Buildings

Syllabus (as given):

Introduction - elastic properties of masonry assemblage - categories of masonry buildings - behaviour of unreinforced and reinforced masonry walls - behaviour of walls - box action and bands - behaviour of infill walls - improving seismic behaviour of masonry buildings - load combinations and permissible stresses - seismic design requirements - lateral load analysis of masonry buildings.

1. INTRODUCTION

Masonry (brick, stone, concrete block, adobe) is the most common construction in India and the developing world - ~ 70% of the housing stock. Masonry is heavy (large inertia force), strong in compression but weak in tension and shear, brittle, and has little energy dissipation. Therefore unreinforced masonry (URM) buildings are the most vulnerable in earthquakes (Latur 1993, Bhuj 2001, Kashmir 2005, Nepal 2015).

- *Typical failure: wall cracking (X-cracks), out-of-plane collapse of walls, separation at corners, roof/gable collapse, falling parapets.*
- *IS codes: IS 1905 (structural use of unreinforced masonry), IS 4326:2013 (earthquake resistant design & construction of buildings), IS 13828:1993 (improving EQ resistance of low-strength masonry buildings), IS 13827 (earthen buildings), IS 1893 (loads), IS 456 (RC bands).*
- *Load-bearing masonry walls serve as both structure and partition, so the layout of walls (symmetry, continuity) controls seismic behaviour.*

Key idea

Basic principle: since masonry cannot be ductile, seismic resistance is obtained by (i) strength (good materials/mortar), (ii) integrity - box action with RC bands and vertical bars, and (iii) regular, simple configuration.

2. ELASTIC PROPERTIES OF MASONRY ASSEMBLAGE

Masonry is a composite of units (brick/block) and mortar - anisotropic, non-homogeneous. Properties depend on unit strength, mortar strength, bond, workmanship, and moisture.

Property	Symbol / value	Remarks
Compressive strength of masonry	f_m (or f'_m) from prism tests	Depends on f_b (unit) and f_{mortar}; IS 1905 gives basic stress tables; $f_m \approx 0.4\text{-}0.6\, f_b$ for typical mortars
Modulus of elasticity	$E_m = 550\, f_m$ (IS 1905); other codes 300-1000 f_m	For brick masonry $E_m \approx$ 2500-6000 MPa; ASCE: 700 f_m (clay), 900 f_m (concrete)

Property	Symbol / value	Remarks
Modulus of rigidity	$G_m = 0.4\ E_m$	Poisson's ratio $\nu \approx 0.25$ (0.15-0.3)
Tensile (flexural) strength	$f_t \approx 0.05$ - $0.15\ f_m$ ($\approx$ 0.1-0.3 MPa)	Depends on bond between mortar and unit; low, variable
Shear strength	$\tau = \tau_0 + \mu\ \sigma_d$ (Coulomb) with $\tau_0 \approx$ 0.1-0.3 MPa, $\mu \approx$ 0.4-0.6	σ_d = compressive stress from dead load; more gravity load -> more shear strength
Unit weight	18-20 kN/m³ (brick), 22-26 (stone)	Heavy -> large seismic force

- Stress-strain curve: non-linear, brittle; peak at strain ≈ 0.002; crushing 0.003-0.0035. Mortar is usually weaker than unit and expands laterally more -> units are in lateral tension, mortar in triaxial compression -> splitting failure of units.
- Masonry strength is directional: strength normal to bed joints differs from parallel (weakest in the stepped joint direction).
- Prism test (h/t ratio 2-5) gives f_m; slenderness and eccentricity reduce permissible stress by factors (stress reduction factor k_s, area factor k_a, shape factor k_p).
- Cracking occurs at very low strain; stiffness degrades after cracking -> effective stiffness ~ 0.5 of uncracked.

3. CATEGORIES OF MASONRY BUILDINGS

IS 4326:2013 classifies masonry buildings into categories (A, B, C, D, E) based on seismic zone and importance factor I - which then decide the type of strengthening required. Higher category = greater strengthening.

Seismic zone factor / Importance	I = 1.0 (ordinary)	I = 1.5 (important)
II (Z = 0.10)	A	B
III (Z = 0.16)	B	C
IV (Z = 0.24)	C	D
V (Z = 0.36)	D	E

Category from zone and importance (concept as per IS 4326; check table in code)

- Category A - minimum strengthening (e.g., in Zone II); E - most stringent.
- Strengthening items depend on category: horizontal bands (plinth, lintel, roof; sill band), vertical reinforcement at corners/T-junctions/jambs, limits on openings, minimum wall thickness, mortar grade, number of storeys.
- Based on construction type: (i) unreinforced load-bearing masonry, (ii) confined masonry (masonry walls with tie-columns and tie-beams cast after walls), (iii) reinforced masonry (steel in cores/horizontal joints), (iv) infilled RC frames (non-structural masonry in frame), (v) stone/rubble masonry, (vi) adobe/earth.
- Number of storeys is restricted by category and masonry type (IS 4326 Table); storey height ≤ 3.4 m; plinth height ≥ 300-450 mm.

Masonry type	Features	Seismic performance
URM (unreinforced)	Bricks + mortar only	Poor; brittle; avoid in high zones without bands
Confined masonry	Tie columns & beams around walls	Good for low-rise (Chile, Mexico); economical
Reinforced masonry	Steel bars in cores/bed joints, grouted	Good; ductile if detailed (R = 3)
Infilled frame	RC/steel frame + masonry infill	Infill increases stiffness/strength; failure modes (see sec. 7)

4. BEHAVIOUR OF UNREINFORCED & REINFORCED MA

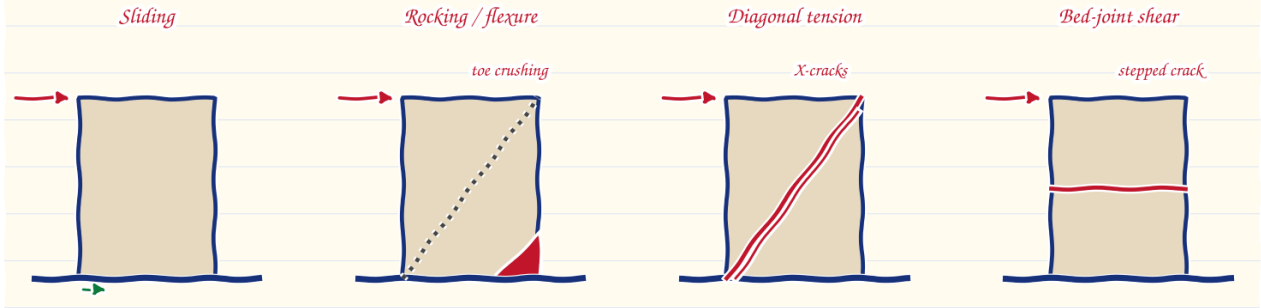


Fig: In-plane failure modes of masonry walls

4.1 In-plane behaviour (shear wall action)

- Bed-joint sliding - horizontal shear crack along a mortar bed joint; occurs at low vertical load / weak mortar.
- Diagonal tension (shear) cracking - X-shaped cracks through units/mortar; most common brittle failure for squat walls with moderate vertical load.
- Rocking (flexure) - slender wall rotates about the toe, tensile cracking at base, toe crushing; ductile compared to shear modes.
- Toe crushing under high vertical load + bending.
- The lateral strength depends on wall length and thickness; the piers between openings act like short columns and fail in shear first. Openings reduce stiffness and strength significantly.

4.2 Out-of-plane behaviour

- Walls perpendicular to ground motion behave as vertical slabs/cantilevers supported at roof and cross-walls; failure by overturning, bending tension cracks mid-height, or horizontal flexural cracking.
- Resistance depends on slenderness ratio (h/t, L/t), edge restraint (cross walls, bands), vertical load; long unsupported walls are most vulnerable (wall length between cross walls ≤ 6 m).
- Gable walls, parapets and cantilevers (chhajjas) fall off first - tie with gable bands and keep parapets ≤ 1 m or reinforce.

4.3 Reinforced masonry

- Vertical bars in cores of hollow blocks or in cavity, horizontal bars in bed joints/bond beams; grout fills the cavities.

- Reinforcement provides tensile and shear capacity, ductility, post-cracking strength and holds fragments together; gives R = 3 (vs 1.5 for URM).
- Typical minimum steel about 0.07% in each direction (total ≈ 0.2%), bars at spacing ≤ 1.2 m, with lap lengths, anchorage and cover as per code (values vary by code).
- Behaviour: flexural cracking -> yielding of vertical steel -> sliding/rocking -> large drift capacity (~ 1-2%) vs ~0.2-0.4% for URM.

Aspect	Unreinforced (URM)	Reinforced masonry
Tension / shear	Almost none (brittle)	Carried by steel
Ductility	Very low, sudden collapse	Moderate; yielding of bars
Drift capacity	0.2 - 0.4%	1 - 2%
R (IS 1893)	1.5	3.0
Post-cracking	Disintegrates, falls	Fragments held together

5. BOX ACTION AND BANDS

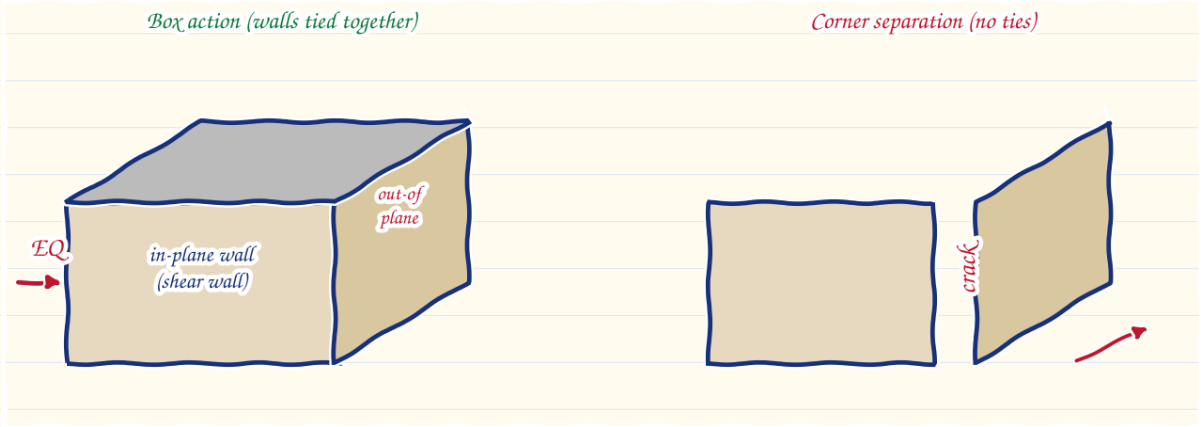


Fig: Box action: walls act together - in-plane walls resist; cross walls hold the out-of-plane walls

- Box action: a masonry building behaves well when all walls and floors/roof are tied together so the building acts like a rigid box - walls parallel to ground motion resist in plane (strong), while walls normal to the motion are supported by them at their ends.
- Without proper connection, corners separate (vertical cracks at corners), walls fall out of plane, roof slides off.
- Requirements: good wall-to-wall bond (toothed joints/stepping), integral horizontal bands, vertical steel at corners/junctions, rigid roof/floor diaphragm (RC slab or tied joists), positive connection of roof to walls.

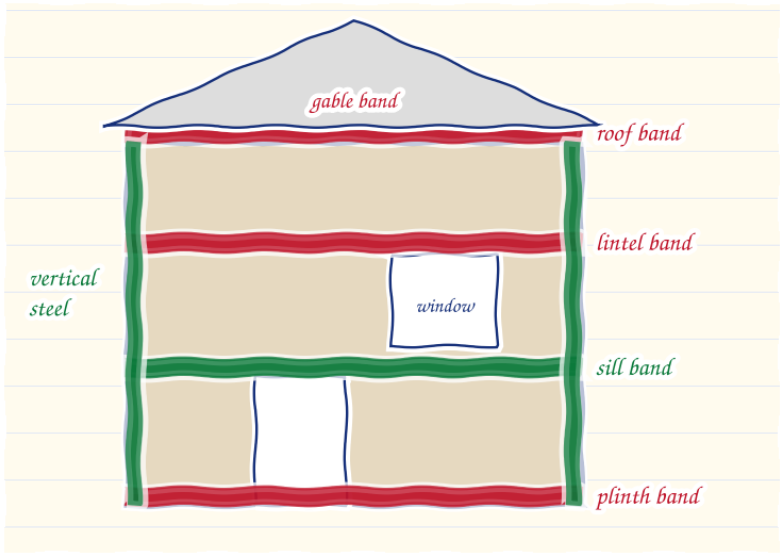


Fig: Location of horizontal seismic bands and vertical steel

Band	Purpose / details
Plinth band	At plinth level in walls on soft/uneven ground; ties walls and reduces differential settlement; not needed if plinth is on firm rock/hard soil with strip footing
Lintel band	At lintel level (all internal and external walls); most important band: ties walls, reduces effective span of walls between cross walls, restrains out-of-plane bending; may be merged with lintels
Roof/floor band	Needed under flexible roofs (timber, sheet) and below RC slab where slab is not properly connected; ensures a continuous tie at the top
Gable band	At the top of gable walls tying them to roof; prevents gable collapse
Sill band	Optional at window sill; in high zones, reduces stress concentration at corners of openings

- RC band: depth ≥ 75 mm (concrete M15/M20 minimum), width = wall thickness; longitudinal steel 2 bars of 8 mm (Category A/B) to 4 bars of 10-12 mm (D/E) with 6 mm stirrups @ 150 mm; continuous around the building; bars properly lapped (bar splices at corners).
- Alternative: timber or steel-bars-in-mortar bands (low-cost buildings, IS 13828).
- Vertical steel at corners, T-junctions, around doors/windows: bar sizes 10-16 mm, anchored in foundation and into roof band; grouted in core of masonry or in "dowels" (increase as category increases).

6. BEHAVIOUR OF INFILL WALLS

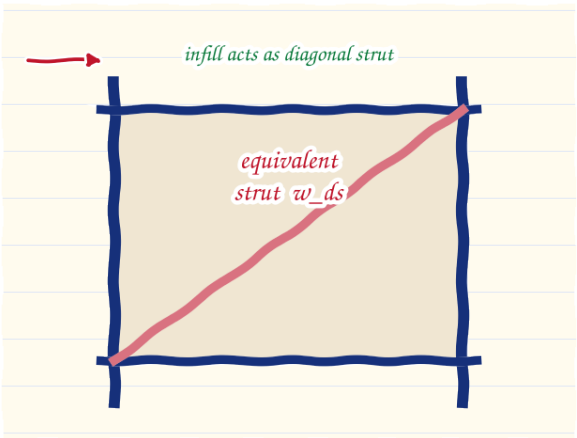


Fig: Infill as an equivalent diagonal strut

- Infill masonry in RC/steel frames is considered non-structural, but it interacts strongly with the frame: it increases stiffness and strength, reduces period and drift initially.
- Under lateral load, frame and infill separate along 2 corners; infill behaves as a compression strut along the diagonal. Equivalent strut width $w_{ds} = 0.175\,\alpha_h^{-0.4}\,L_{ds}$ (IS 1893:2016 cl 7.9.2.2), where $\alpha_h = h\,[E_m\,t\,\sin 2\theta/(4\,E_f\,I_c\,h_{inf})]^{1/4}$; L_{ds} = diagonal length; thickness = infill thickness.
- Positive: extra stiffness/strength/energy dissipation (friction, cracking) - improves performance if present uniformly.
- Negative: (a) infill fails in brittle manner (diagonal cracking, corner crushing); (b) soft storey where ground storey has no infill; (c) short column by partial infill; (d) torsion from asymmetric infill; (e) out-of-plane collapse of slender infill walls; (f) strut action induces high shear in columns -> shear failure.
- Design implications (IS 1893:2016): model infill as struts when stiffness matters; use $T_a = 0.09h/\sqrt{d}$; provide gaps / flexible joints (isolated infill) if frame performance is to be independent; tie infill with dowels/mesh and bands for out-of-plane; avoid openings near columns; ground storey with infill in parking to avoid soft storey.

7. IMPROVING SEISMIC BEHAVIOUR OF MASONRY BUILI

Issue	Provision (IS 4326 / 13828)
Configuration	Simple rectangular plan; L/B ≤ 3; symmetrical; plinth/roof levels uniform; avoid large re-entrant corners; use separate blocks (seismic joint) for L/T plans
Material	Brick strength ≥ 3.5 N/mm² (35 kg/cm²); mortar cement : sand = 1:6 (Cat A,B) to 1:4 (D,E); lime-cement mortar; avoid mud mortar in high zones
Wall thickness & height	t ≥ 200 mm (230 mm brick); storey height ≤ 3.4 m; height/thickness ≤ 12-15; length of wall between cross walls ≤ 6 m
Openings	Total opening length ≤ 50% (1-storey), 42% (2-storey), 33% (3-storey) of wall length; distance from corner ≥ 600 mm; pier width ≥ 600 mm; opening must be centrally located and symmetrical
Bands & steel	Provide plinth, lintel, roof, gable bands; vertical steel at corners & jambs (increase with category)
Bonding	Toothed (stepped) joints at wall intersections; full bond; avoid vertical straight joints; header courses

Issue	Provision (IS 4326 / 13828)
Roof / floor	RC slab preferable; timber/GI roofs braced; tie rafters; roof must act as a rigid diaphragm; no heavy parapets/water tanks on roof
Foundations	Same type/level throughout; strip footings; plinth beam if soil variation
Non-structural	Parapets ≤ 1 m (reinforced), chhajjas ≤ 0.6-1 m; chimneys braced; tie water tanks; avoid cantilevers
Quality	Good workmanship; curing; no loose/eccentric loads

Strengthening

Retrofitting options for existing masonry: ferrocement/mesh jacketing (shotcrete), RC jacketing, horizontal & vertical steel bands (external), grouting of cracks, tying walls (steel rods) at floor level, enlarging piers, adding RC wall/buttress.

8. LOAD COMBINATIONS & PERMISSIBLE STRESSES

- Load combinations (working stress method for masonry as per IS 1905/4326): (i) DL + IL ; (ii) DL + IL ± EL ; (iii) DL ± EL (for overturning/uplift check) ; wind not combined with EQ. IL taken as per IS 1893 (25% / 50% in seismic weight, 0% on roof).
- Permissible stresses under EQ load combination are increased by 33.33% (1/3) over basic values; permissible tensile and shear values depend on mortar and unit strength.
- Basic compressive stress (IS 1905 table) is multiplied by stress-reduction factors for slenderness/eccentricity (k_s), area (k_a) and shape (k_p).
- Permissible shear stress: $\tau_p = \tau_0 + 0.1\,\sigma_d$ (σ_d = dead load stress) - basic shear for mortar M1 (1:6) ~0.05-0.1 N/mm²; min tensile bond ~0.05-0.1 MPa.
- Tension is generally not allowed in URM except small values; where tension develops (overturning), reinforcement must resist it.

9. SEISMIC DESIGN REQUIREMENTS (IS 4326 / 1893)

- Seismic weight W = DL + 25% IL (IL ≤ 3 kN/m²) or 50% IL; roof IL = 0. Seismic zone factor, I, R = 1.5 (URM) to 3.0 (confined/reinforced).
- Design horizontal seismic coefficient $A_h = (Z/2)(S_d/g)/(R/I)$, with S_d/g from IS 1893 spectrum (time period for masonry ≈ 0.09 $h/\sqrt{d}$, usually small, so $S_d/g = 2.5$).
- For low-rise masonry the design coefficient is taken directly from IS 1893 (IS 4326 refers to it); in the older seismic coefficient method $A_h = \beta\, I\, \alpha_0$.
- Vertical projections (parapets, balconies, chhajjas) are designed for about 5 times A_h acting on their weight (IS 1893).
- Walls must be checked for in-plane shear, in-plane flexure, out-of-plane bending (as vertical strip supported at floor/roof levels) and overturning.
- Number of storeys is limited by category and masonry type (IS 4326 Table) - fewer storeys in higher zones, stone/earthen buildings lowest.

10. LATERAL LOAD ANALYSIS OF MASONRY BUILDINGS

1. Compute seismic weight of each floor (roof) including half of wall weights above and below.
2. Base shear $V_B = A_h W$; distribute with height $Q_i = V_B W_i h_i^2 / \Sigma W_j h_j^2$ (for low-rise often triangular).
3. Distribute floor force to walls (in each direction): rigid diaphragm (RC slab) -> in proportion to rigidity (relative stiffness) of walls; flexible diaphragm -> in proportion to tributary mass / area.
4. Add torsion (CM, CR) with accidental eccentricity 0.05 b.
5. Compute shear force in each wall/pier and shear stress $\tau = V/(L\,t)$; compute bending stress and axial stress; check against permissible (+33%).
6. Check overturning / uplift of walls using 0.9 DL.

Wall rigidity (cantilever vs fixed-fixed)

Lateral stiffness of a wall (shear + flexure): $k = 1 / (\Delta \text{ per unit load}) = 1/[\, h^3/(3 E_m I) + 1.2\, h/(G_m A)\,]$ (cantilever)

Fixed-fixed (pier between openings, restrained top & bottom): $\Delta = h^3/(12 E_m I) + 1.2\, h/(G_m A)$

Squat walls ($h/L < 0.4$): shear deformation dominates, $k \approx G_m A/(1.2\, h) = G_m\, t\, L/(1.2\, h)$; slender walls: flexure dominates, $k \propto t\, L^3/h^3$.

Relative rigidity of a wall = $k_{wall} / \Sigma k$ (used for distributing floor shear).

Solved example - single-storey brick building

Data

Single storey, plan 8 m x 6 m, walls 230 mm thick, height 3.0 m, RC slab roof. Zone IV ($Z = 0.24$), $I = 1.0$, $R = 1.5$ (URM), medium soil; walls 19 kN/m³.

Weights: roof slab (150 mm + finish) 5.5 kN/m² x 48 = 264 kN; walls (perimeter 28 m + 1 internal wall 6 m = 34 m): 0.23 x 3.0 x 34 x 19 = 446 kN, half = 223 kN

Seismic weight of roof level: $W = 264 + 223 = 487$ kN (roof LL ignored); $T_a = 0.09 \times 3.0/\sqrt{6} = 0.11$ s -> $S_d/g = 2.5$

$A_h = (0.24/2) \times 2.5/(1.5/1.0) = 0.12 \times 2.5/1.5 = 0.20$; $V_B = 0.20 \times 487 = 97.4$ kN

Distribution to walls in the 6-m direction (loading along 8 m walls)

Two 8 m long walls (no openings) take the shear; assume each rigid by length (squat): V per wall = 97.4/2 = 48.7 kN

Shear stress $\tau = V/(L\,t) = 48.7 \times 10^3/(8000 \times 230) = 0.026$ N/mm²

Permissible shear (mortar 1:6) $\tau_p = 0.05\text{-}0.1 + 0.1\,\sigma_d$; σ_d (roof + wall) ≈ 0.12 N/mm² -> $\tau_p \approx 0.07$ N/mm² x 1.33 = 0.093 N/mm² -> 0.026 < 0.093 OK

Out-of-plane: 6-m wall spanning 8 m wall-to-wall - verify $h/t = 3000/230 = 13$ (limit ~ 15) and add lintel band + vertical steel at corners.

11. MORE WORKED EXAMPLES

11.1 Relative rigidity of walls with openings

Wall with two piers (cantilever method)

Wall height $h = 3.0$ m, thickness $t = 0.23$ m, $E_m = 2750$ N/mm², $G_m = 0.4\,E_m = 1100$ N/mm². Piers: P1 length 1.2 m, P2 length 2.4 m.

Cantilever stiffness: $k = 1/[h^3/(3E_m I) + 1.2h/(G_m A)]$ with $I = t L^3/12$, $A = tL$

P1: $I = 0.23 \times 1.2^3/12 = 0.0331$ m⁴ -> flexure $27/(3 \times 2.75e6 \times 0.0331) = 9.9e\text{-}5$ m/kN; shear $3.6/(1.1e6 \times 0.276) = 1.19e\text{-}5$ -> $\Delta = 1.11e\text{-}4$ -> $k_1 = 9000$ kN/m

P2: $I = 0.23 \times 2.4^3/12 = 0.265$ m⁴ -> flexure $27/(3 \times 2.75e6 \times 0.265) = 1.23e\text{-}5$; shear $3.6/(1.1e6 \times 0.552) = 5.9e\text{-}6$ -> $\Delta = 1.82e\text{-}5$ -> $k_2 = 54\,900$ kN/m

Share of storey shear: P1 = 9000/63 900 = 14%, P2 = 86% -> longer pier attracts most force; narrow piers rely on bands/steel.

Learning

Observation: stiffness $\propto L^3$ for slender walls but $\propto L$ for squat walls. Door/window openings cut the wall into piers and reduce lateral strength drastically - hence opening limits (50%/42%/33%).

11.2 Overturning check

Check wall stability

Wall 4 m long, 3 m high, $t = 0.23$ m, with roof load 30 kN/m on top; base shear force at roof $V = 40$ kN (A_h W portion).

Weight $W = 0.23 \times 3 \times 4 \times 19 = 52.4$ kN + roof $30 \times 4 = 120$ kN -> $N = 172.4$ kN; use 0.9 DL = 155 kN for overturning

Overturning moment $M = 40 \times 3 = 120$ kN.m; restoring moment $= 155 \times 4/2 = 310$ kN.m -> F.S. = 2.6 > 1.5 OK

Base stress $\sigma = N/A \pm M/Z = 155/(0.23x4) \pm 120/(0.23 \times 4^2/6) = 168 \pm 196$ (kN/m²) -> tension 28 kN/m² at heel: small tension -> reinforce vertical bars / increase dead load.

11.3 Out-of-plane check of a wall strip

- Treat a 1 m strip as vertical slab supported at base and roof; seismic pressure $p = A_h \times w$ (w = weight per m² of wall = $0.23 \times 19 = 4.4$ kN/m²).
- For $A_h = 0.20$: $p = 0.88$ kN/m²; moment (propped cantilever / simply supported) $M = p\,h^2/8 = 0.88 \times 9/8 = 0.99$ kN.m/m
- Section modulus $Z = 1 \times 0.23^2/6 = 8.8 \times 10^{-3}$ m³ -> bending stress $\sigma = 0.99/8.8e\text{-}3 = 112$ kN/m² (0.11 MPa) vs permissible tensile strength of masonry about 0.05-0.1 MPa -> cracks; need vertical steel/band or reduce slenderness.

12. STONE, EARTHEN AND LOW-STRENGTH MASONRY

- Stone masonry: heavy, irregular, weak mortar (mud/lime) -> poor performance (Latur 1993). Needs through-stones (one per 600 mm), horizontal bands (timber/RC), vertical bars at corners (IS

4020, IS 13828). Limit: height ≤ 1-2 storeys, thickness ≥ 450 mm.

- Adobe / earth buildings (IS 13827): very heavy and weak; strengthened by bamboo/steel mesh, wall-corner reinforcement, light roof (reduce mass), buttresses, strong plinth.
- Low-strength masonry (IS 13828): brick in mud mortar - provide rope-like bands at plinth, lintel, roof; vertical bars at corners; roof tied by a ring beam; use light roof (GI sheet/tiles instead of heavy earth roof).
- Common local techniques: bamboo reinforcement, dhajji dewari (timber frame with infill - Kashmir), Bhatar (Himachal timber-laced stone masonry).

Confined masonry (modern low-rise practice)

- Masonry walls are built first; then RC tie-columns at corners/jambs and tie-beams at floor levels are cast against the walls (toothed interface). Walls carry the vertical load, the frame confines them.
- Confining elements prevent wall disintegration after cracking, raise drift capacity (~ 0.5-1%), and give box action; limits: tie-columns ≥ 150 mm (typ. 4 bars 10 mm), tie-beam at each floor, spacing of columns ≤ 4-5 m.

13. LESSONS FROM PAST EARTHQUAKES - MASONRY

Earthquake	Observed masonry damage	Lesson
Latur 1993 (M 6.2)	Stone masonry with heavy mud roofs collapsed; ~8000 deaths	Light roof, bands, vertical steel, avoid mud mortar
Bhuj 2001 (M 7.7)	Brick/stone houses collapsed, corner separation, parapets and gables fell	Box action, bands, quality, anchorage of parapets
Kashmir 2005 (M 7.6)	Stone masonry with heavy roofs; dhajji dewari survived better	Light timber-braced systems good
Nepal 2015 (M 7.8)	Mud-mortar stone & brick URM; temples; infilled RC soft-storey	Retrofitting and training of masons
Sikkim 2011	Infill and masonry damage, poor detailing	Seismic code compliance

Common deficiencies in masonry buildings

- Poor mortar and workmanship; no bands; unbraced long walls; large openings near corners; heavy RC/earth roofs on weak walls; shallow/unequal foundations; unreinforced parapets and gables; cut-out walls for services.
- Inadequate roof-to-wall connection - roof slides off walls; floor joists with no anchorage.

FORMULA SHEET - UNIT IV

Item	Formula / value
Elastic modulus	$E_m = 550\, f_m$; $G_m = 0.4\, E_m$; $\nu \approx 0.25$
Shear strength	$\tau = \tau_0 + \mu\, \sigma_d$ (permissible: $\tau_0 + 0.1\, \sigma_d$, $\times 1.33$ for EQ)
Design coefficient	$A_h = (Z/2)(S_a/g)/(R/I)$; $R = 1.5$ (URM), 3.0 (reinforced)

Item	Formula / value
Base shear and distribution	$V_B = A_h W$; $Q_i = V_B W_i h_i^2 / \Sigma W h^2$
Wall stiffness	$k = 1/[h^3/(3E_m I) + 1.2h/(G_m A)]$ (cantilever)
Shear stress in wall	$\tau = V/(L t)$
Openings limit	≤ 50% / 42% / 33% (1 / 2 / 3 storeys); corner distance ≥ 600 mm
Infill strut	$w_{ds} = 0.175 \alpha_h^{-0.4} L_{ds}$

SHORT ANSWERS (2-mark / viva)

Question	Answer
Why is masonry vulnerable?	Heavy (high inertia), brittle, very low tensile & shear strength, little ductility, poor connections
What is box action?	Walls, floors and roof tied so that the building responds as a single rigid box; in-plane walls resist, cross walls support out-of-plane walls
Which band is most important?	Lintel band (ties all walls, reduces out-of-plane bending)
Role of vertical steel?	Resists vertical bending/rocking, holds corners, increases ductility and prevents collapse of wall pieces
Max length/height of unsupported wall?	Wall length between cross walls ≤ 6 m; storey height ≤ 3.4 m; h/t ≤ 12-15
Why limit openings?	Openings weaken wall; piers fail in shear; limit 50% / 42% / 33% of wall length
Infill: advantage and disadvantage?	Adds stiffness/strength; but creates soft storey, short column, torsion; brittle failure
How is mortar strength specified?	By cement : sand ratio (1:6 to 1:4) or compressive strength grades M1-M3 (IS 1905)
R for URM / reinforced masonry?	1.5 / 3.0 (IS 1893:2016)
How to improve out-of-plane strength?	Reduce slenderness, add cross walls/bands/buttresses, vertical steel, tie wall to roof

Checklist for a seismic-safe masonry house (IS 4326)

1. Simple rectangular plan, L/B ≤ 3, symmetric; separate blocks if irregular.
2. Good bricks/stones, richer mortar, full bond with toothed corners.
3. RC plinth band (if needed), lintel band, roof band, gable band - continuous & lapped properly.
4. Vertical bars at all corners/junctions/jambs, anchored in foundation and roof band.
5. Openings small, central and ≥ 600 mm from corners; total ≤ 50% (1 storey).
6. Light roof / RC slab tied to walls; no heavy parapets/overhangs/water tanks on top.
7. Uniform foundation level and type; good drainage; quality workmanship and curing.

QUICK REVISION - Likely questions

1. Elastic properties of masonry: E_m, G_m, strength, stress-strain behaviour.
2. Categories of masonry buildings (A-E) as per IS 4326; strengthening required.
3. Behaviour of URM and reinforced masonry walls in-plane and out-of-plane; failure modes.
4. Explain box action; role of horizontal bands and vertical reinforcement (sketch).
5. Behaviour of infill walls, equivalent strut, positive and negative effects.
6. Measures to improve seismic behaviour of masonry buildings (IS 4326 provisions).
7. Load combinations/permissible stresses; steps of lateral load analysis; numerical on base shear and wall shear.

UNIT - V

Structural Walls, Non-Structural Elements & Ductility

Syllabus (as given):

Strategies in location of structural walls - sectional shapes - variations in elevation - cantilever walls without openings - failure mechanism of non-structures - effects of non-structural elements on structural system - analysis of non-structural elements - prevention of non-structural damage. Ductility: impact, requirements, assessment, factors affecting ductility - ductile detailing as per IS 13920:2016 - behaviour of beams, columns and joints in RC buildings during earthquakes.

PART A : STRUCTURAL WALLS (SHEAR WALLS)

Definition - Structural (shear) wall

A vertical RC wall designed to resist lateral forces in its own plane (shear, bending, axial) and to give the building stiffness, strength and drift control. Acts as a vertical cantilever fixed at the foundation.

- *Advantages: very high stiffness -> small drift -> less non-structural damage; large strength; efficient in tall buildings (10-40 storeys); can be ductile if detailed (IS 13920).*
- *Shear-wall buildings performed well in Chile 1960/2010 (but needs proper detailing: 2010 boundary-element failures), Bhuj 2001 and Mexico.*
- *Disadvantages: concentrate large forces on foundations (overturning), reduce architectural flexibility.*

1.1 Strategies in the location of structural walls

- *Place walls symmetrically about both axes in plan so that centre of rigidity ≈ centre of mass (minimise torsion).*
- *Locate walls along the periphery / as far from CR as possible to maximise torsional resistance; avoid a single core at one end.*
- *Distribute walls evenly in both orthogonal directions (at least 2 walls in each direction, ideally near the outer edges) - walls should not all be on one side.*
- *Walls must be continuous to the foundation (no discontinuity/transfer girders), with uniform thickness variation in elevation.*
- *Put walls in continuous lines with columns so that slab spans remain small; avoid walls along the weak axis only.*
- *Do not put lift/stair cores at one corner; if unavoidable, add balancing walls on the opposite side.*
- *Walls with openings: openings should be small, aligned vertically, away from the wall ends (boundary zones) - use coupling beams for large vertical rows of openings.*
- *Shear walls at ends of long plans are not advisable: they restrain thermal/shrinkage movement and attract cracking.*

- Provide adequate foundation for walls: large footing/raft/piles to resist overturning (tension at heel).

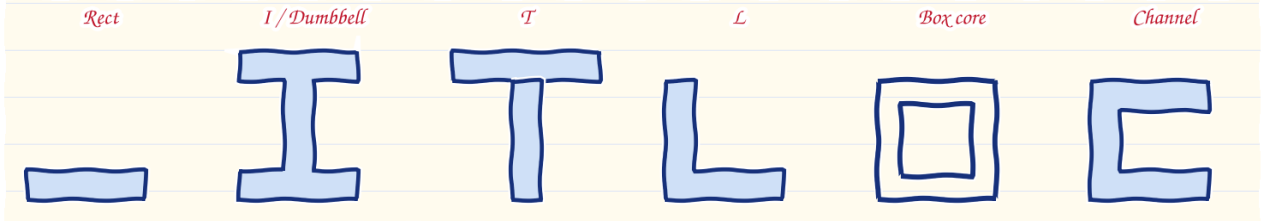


Fig: Sectional shapes of shear walls

1.2 Sectional shapes

Shape	Characteristics
Rectangular	Simplest; strong in own plane only, weak out-of-plane; used with columns as dual system
Barbell / I / dumbbell (flanged)	Boundary columns at wall ends act as flanges - increases flexural strength & stability (preferred where wall carries big moment)
T and L (flanged walls)	Flanges increase bending capacity/stiffness (effective flange width limited by half the distance to the next wall and a fraction of wall height, per code); torsionally unsymmetrical - careful
Channel / U-shaped	Used for lift/stair cores; open sections have low torsional stiffness, shear centre outside section
Box / closed core	Closed core has huge torsional and flexural stiffness; best for tall buildings; openings (doors) reduce stiffness; use coupling beams

1.3 Variations in elevation

- Wall thickness and reinforcement may be reduced gradually with height (shear and moment diminish upward) - avoid sudden changes (strength irregularity).
- Discontinuing a wall at a mid-level (floating wall) makes a weak/soft storey and large forces on transfer beams - prohibited by IS 1893:2016 for tall buildings without special design.
- Openings should occur in vertical rows - coupled shear walls (coupling beams) act as large frames: two walls + coupling beams -> energy dissipation via beam yielding (preferred over weak walls).
- Avoid large set-backs of wall in elevation; vary only gradually.
- Plastic hinge region at the base of wall (height l_w or $h_w/6$, whichever is more) is designed ductile; above it elastic.

1.4 Cantilever walls without openings

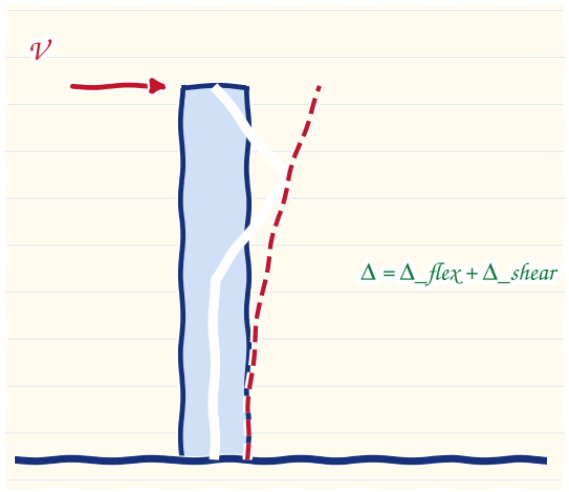


Fig: Cantilever wall: flexure + shear deflection

- Behaves as a vertical cantilever fixed at base: bending moment $M = \Sigma Q_i h_i$ (max at base), shear $V = \Sigma Q_i$ (max at base); axial load N from gravity.
- Slenderness (aspect ratio h_w/l_w): slender (> 2) -> flexure dominated, ductile if steel yields; squat (< 1) -> shear dominated, brittle (diagonal tension/sliding).
- Deflection at top: $\Delta = \Delta_{flex} + \Delta_{shear}$; for a tip load V: $\Delta = V h^3/(3EI) + 1.2\, V h/(GA)$ (for a triangular distribution of lateral load the flexural part is $\Delta_{flex} = 11\, V_B h^3/(60\, EI)$); $G = 0.4E$.
- Strength design: flexural design with P-M interaction (boundary elements), shear design with $V_u \le \tau_c, max$; ensure flexural capacity < shear capacity (capacity design): $V_u = V_{flex} = M_u/h_{eff}$ (× amplification).
- Modes of failure: flexural yielding (ductile), diagonal tension, diagonal compression (web crushing), sliding shear at construction joint, rocking/overturning, boundary-element crushing/buckling.

IS 13920:2016 - Ductile shear wall provisions (summary)

Thickness ≥ 150 mm; reinforcement ≥ 0.25% of gross area, both horizontally and vertically; if t > 200 mm or $V_u > 0.25\sqrt{f_{ck}}\, A_{cv}$ -> two curtains (layers) of steel.

Bar diameter ≤ t/10; spacing ≤ 450 mm (or 3t / l_w/5, least) both ways; vertical bars continue full height; horizontal bars anchored in boundary elements with 135° hook.

Boundary elements required when extreme-fibre compressive stress due to factored loads exceeds $0.2\, f_{ck}$; may be discontinued where stress < $0.15\, f_{ck}$. Boundary element: confining hoops as in columns (spacing ≤ 100 mm), length ≥ $0.15\, l_w$ (and ≥ wall thickness × 1.5), vertical steel 0.8% - 4%.

Shear: nominal shear stress $\tau_v = V_u/(t\, d_w) \le 0.63\sqrt{f_{ck}}$ (max); design shear is increased above analysis shear (capacity design / dynamic amplification for tall walls) so that flexure yields before shear fails.

Coupling beams ($l/d \le 4$): diagonal reinforcement cages in each direction, confined by ties.

Rule of thumb

Practical thumb rule: total shear-wall area in each direction ≈ 1.5% - 2.5% of floor area is enough in zones III-V; walls should be spread; dual systems (walls + SMRF) preferred.

PART B : NON-STRUCTURAL ELEMENTS (NSEs)

Definition - Non-structural elements

Components of a building that do not form part of the load-carrying system: partitions/infill, parapets, ceilings, facade/cladding, glazing, stairs (if isolated), mechanical-electrical-plumbing (MEP) equipment, water tanks, lifts, furniture, contents.

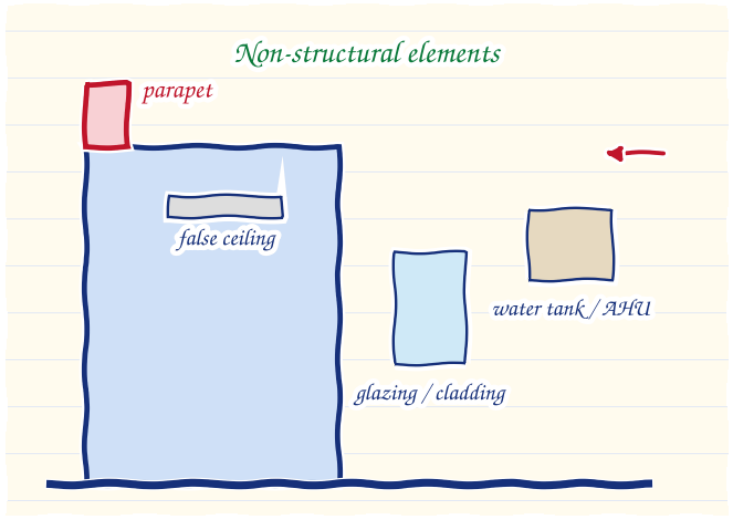


Fig: Typical non-structural elements

Significance: NSEs account for 70-85% of building cost (hospitals, offices) and cause most financial loss and injury (falling hazards), disruption (post-earthquake functionality). Even when the structure survives, NS damage can render a building unusable.

2.1 Failure mechanisms of non-structural elements

Element	Failure mechanism (cause)
Brick infill / partitions	In-plane: diagonal cracking due to storey drift; out-of-plane: overturning/bending - falling of walls
Parapets, chimneys, gables, balconies	Cantilever whiplash amplification at roof; overturning at base
Ceilings, light fittings	Loss of support, suspended grid failure; pounding with walls
Glazing / cladding	Drift imposed on rigid panels -> cracking/crushing; connection failure; falling glass
Heavy equipment/tanks/AHU	Sliding, overturning, loss of anchor; pipe rupture; leakage; fire
Stairs & lifts	Stairs act as diagonal braces -> stiffening -> failure; lift counterweight derailment; rails distortion
Contents / racks / furniture	Overturning, falling objects (books, servers)
Services (pipes, ducts, cables)	Fatigue, rupture at joints crossing seismic gaps

Two causes: (i) acceleration-sensitive (ceilings, equipment, parapets) - damage depends on floor acceleration; (ii) drift-sensitive (partitions, glazing, cladding, stairs) - depends on inter-storey drift.

2.2 Effects of non-structural elements on the structural system

- *Stiffening and strengthening: masonry infill raises stiffness 5-10x, reduces period (higher seismic force) and changes load path - may help (reduce drift) or hurt.*
- *Soft storey when infill is omitted in one floor (parking/entrance); short-column effect with partial infill; torsion from asymmetric infill (e.g., on 2 sides only).*
- *Added mass (heavy parapets, water tanks, partitions) increases base shear and top-heavy response; heavy roof tanks cause whiplash.*
- *Interference with structural detailing: rigid cladding/stairs hindering drift -> unintended forces on columns.*
- *IS 1893:2016: NS elements shall be accounted for in mass and stiffness (as struts) in analysis; separate them where they should not participate (gaps).*

2.3 Analysis of non-structural elements

Design lateral force on NS components/parts (IS 1893:2016 cl. 7.12 - concept)

$F_p = (Z/2)\,(I_p/R_p)\,a_p\,(1 + 2\,z/h)\,W_p$ (general form, as in most codes) - force increases with height z/h above base (up to 3 times at roof)

W_p = weight of the component; a_p = component amplification factor; R_p = component response modification factor; I_p = importance of component (1.0 / 1.5 for hazardous/essential)

Equivalent static force applied at the centre of mass of the component in the most unfavourable direction (horizontal), plus vertical.

- *Always use the factors a_p, R_p given in IS 1893:2016 Cl. 7.12 for the particular component - note them from the code tables in the exam; the idea is floor acceleration amplification x component weight.*
- *Vertical projections (parapets, balconies, canopies) are designed for about 5 times A_h on their weight (IS 1893).*
- *Drift check for drift-sensitive elements: relative displacement D_p = (drift ratio) x h_{storey}; glazing needs clearance $\delta \geq 1.25\,D_p$ (+ some); partition gaps ≥ computed drift.*
- *Equipment: designed to resist F_p, anchors (bolts, cast-in plates) checked for tension/shear with factor ≥ 1.0-1.5; flexible connections for piping.*

Worked example - parapet

Parapet 1.0 m high, 115 mm brick: weight per metre length W_p = 0.115 x 1.0 x 19 = 2.19 kN/m; building A_h = 0.10 (Zone IV); projection rule $F_p = 5\,A_h\,W_p$.

F_p = 5 x 0.10 x 2.19 = 1.09 kN/m acting at 0.5 m above base -> overturning moment = 1.09 x 0.5 = 0.55 kN.m/m

Resisting moment of self weight = 2.19 x 0.115/2 = 0.126 kN.m/m -> 0.126 < 0.55: an unreinforced parapet overturns.

Remedy: RC parapet / vertical bars anchored into slab, reduce height to ≤ 0.6 m, or brace.

2.4 Prevention of non-structural damage

- *Isolate drift-sensitive NSEs: gaps/sliding joints between infill and frame (isolated infill), flexible sealants, glazing with clearance, partition tracks.*
- *Anchor/brace acceleration-sensitive items: parapets (reinforced, ≤ 1 m), water tanks, AHUs, cable trays, suspended ceilings (diagonal bracing wires), shelving to walls.*
- *Reduce mass and keep it low; avoid heavy elements at the top.*
- *Positive connections: dowels/mesh to tie masonry infill to frame; bands in infill; anchors for cladding with slotted holes.*
- *Flexible service connections across seismic joints; shut-off valves; battery lighting; fire-stop.*
- *Limit drift to 0.004 h (IS 1893) to avoid NS damage; use stiffer structure for hospitals (I = 1.5).*
- *Quality control + regular inspection; code compliance for MEP (IS 1893 Part 1 cl 7.12; FEMA 74 guidelines).*

PART C : DUCTILITY CONSIDERATIONS IN EQ RESISTANT

3.1 Introduction and impact of ductility

- *A ductile structure deforms inelastically without strength loss; the area under the force-displacement loop = energy dissipated.*
- *Design philosophy: code forces are 1/R of elastic forces; the structure survives MCE only because of ductility (and over-strength).*
- *Impact of ductility: (i) reduces design force (R), (ii) prevents sudden collapse, (iii) redistributes moments, (iv) gives warning before failure (large cracks/deflection), (v) energy dissipation through hysteresis, (vi) allows structure to survive even when ground motion exceeds the design level.*

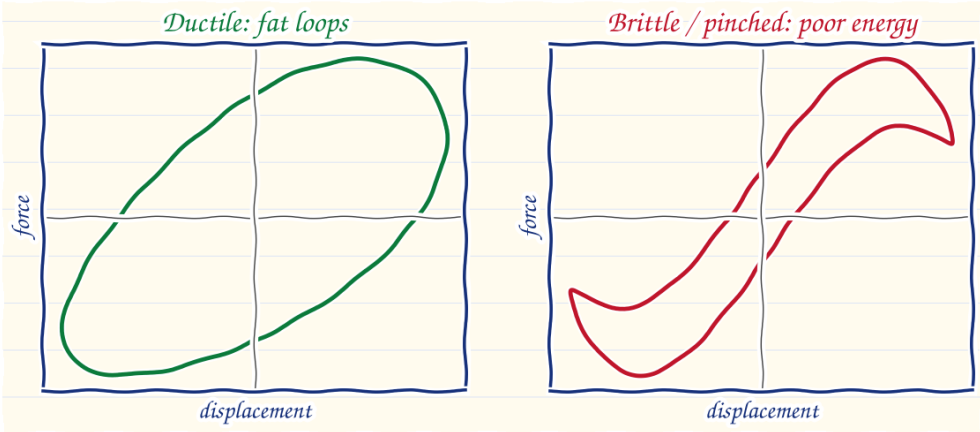


Fig: Hysteresis loops: ductile (fat) vs brittle (pinched)

3.2 Requirements for ductility

- *Material: Fe 415/500 (D) steel with elongation ≥ 14.5% and UTS/YS ≥ 1.15; concrete ≥ M20 (M25 in tall buildings, Zone III-V).*
- *Member level: under-reinforced beams (tension steel yields first), limited axial load ratio in columns ($P/(f_{ck} A_g) \leq 0.4$-0.5), confined concrete, adequate shear capacity.*
- *System level: weak beam-strong column ($\Sigma M_c \geq 1.4\, \Sigma M_b$), regular configuration, redundancy, no soft storey, ductile joints.*

- *Detailing: hooks 135°, closely spaced stirrups, adequate lap/anchorage, no splicing in plastic hinge zones.*
- *Plastic hinge rotation capacity: $\theta_p = (\varphi_u - \varphi_y) L_p$; $L_p \approx 0.5\,d + 0.05\,z$ (Paulay-Priestley) or $\approx D$.*

3.3 Assessment of ductility

Type	Definition	Typical
Displacement ductility	$\mu_\Delta = \Delta_u / \Delta_y$ (from pushover / load-deflection)	SMRF: 3-5
Curvature ductility	$\mu_\varphi = \varphi_u / \varphi_y$ (moment-curvature M-φ of section)	10-20 for good beam sections
Rotation ductility	$\mu_\theta = \theta_u / \theta_y$	related to hinge
Energy ductility	$E_{dissipated} / E_{elastic}$	Hysteresis area
Overall relation	$\mu_\Delta = 1 + 3(\mu_\varphi - 1)(L_p/L)(1 - 0.5\,L_p/L)$	Cantilever of length L

Moment-curvature quantities

$\varphi_y = \varepsilon_y / (d - x_u, y)$ ($\approx 2.1\,\varepsilon_y / D$ for beams); $\varphi_u = \varepsilon_{cu} / x_u = 0.0035 / x_u$ (IS 456 limit strain)
$\mu_\varphi = \varphi_u / \varphi_y$; tension steel increases x_u -> lowers μ_φ; compression steel and confinement raise μ_φ

Numerical

Example: beam with $\varphi_y = 5.5 \times 10^{-6}$ /mm and $\varphi_u = 5.0 \times 10^{-5}$ /mm -> $\mu_\varphi = 9.1$; with $L_p/L = 0.1$: $\mu_\Delta = 1 + 3(8.1)(0.1)(0.95) = 3.3$.

3.4 Factors affecting ductility

Factor	Effect on ductility
Tension steel ratio ρ	Increase in ρ reduces ductility (x_u rises; $\rho_{max} = 2.5\%$ in beams)
Compression steel ρ′	Increases ductility (reduces x_u); IS 13920: A_s,bottom ≥ 50% A_s,top at joint face
Axial load ratio	Higher axial load reduces ductility - limit $P/(f_{ck} A)$ in columns
Confinement (hoops, spacing, hook)	Closer ties & 135° hooks increase ductility strongly
Concrete strength	Higher f_{ck} -> more brittle (lower ε_{cu} in unconfined) - use M20-M30; avoid very high grades
Steel grade / ε_y	Lower yield strain (lower f_y) improves μ; strain-hardening ratio > 1.15 ideal; avoid CTD bars
Shear span / shear stress	Low shear span ratio (short members) -> brittle shear failure
Bond/anchorage and lap	Poor bond/lap in hinge region reduces ductility - use hooks & lap outside hinge zones
Cover and bar spacing	Spalling of cover removes confinement - small cover, tight cage

PART D : DUCTILE DETAILING AS PER IS 13920:2016

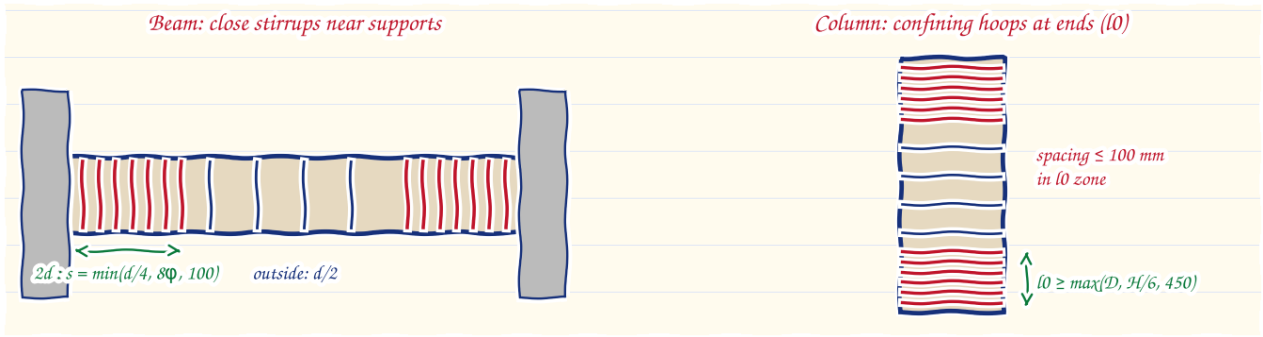


Fig: Confining reinforcement at member ends

4.1 Beams

- Geometry: width $b \geq 200$ mm; $b/D \geq 0.3$; depth $D \leq$ 1/4 clear span; .
- Longitudinal steel: min $\rho = 0.24\sqrt{f_{ck}}/f_y$; max $\rho = 2.5\%$; at least 2 bars at top and bottom through the full length; positive steel at joint face ≥ 50% of negative steel there; any section top/bottom ≥ 25% of max moment capacity at the joint face.
- Lap splices: only when hoops are provided over the lap length (spacing ≤ 150 mm); not within a joint, nor within 2d from joint face, nor at locations of yield; ≤ 50% bars lapped at one section.
- Web (stirrup) reinforcement: closed hoops with 135° hook, extension ≥ 10 d_b (≥ 75 mm); first hoop at 50 mm from the joint face; spacing in the hinge zone (2d from face) = min(d/4, 8 d_b, 100 mm); beyond that ≤ d/2; diameter ≥ 6 mm (8 mm if clear span > 5 m).
- Design shear: $V_e = V_{u,gravity} \pm 1.4\,(M_{u,lim}^{sag} + M_{u,lim}^{hog})/L$ (capacity design) - ensures flexural failure first.

4.2 Columns

- Geometry: min dimension ≥ 300 mm; ratio of shorter to longer side $b/D \geq 0.4$.
- Longitudinal steel: min 0.8% and max 4% of gross area (6% not allowed in lapped region); bars spaced ≤ 200 mm; lap splices in the central half of column height, with hoop spacing ≤ 150 mm over lap.
- Confining hoops at both ends over length $l_0 \geq \max(D, H_{clear}/6, 450\text{ mm})$ - spacing $s \leq$ 1/4 of min member dimension, and 75 mm ≤ s ≤ 100 mm; dia ≥ 8 mm; hoops with 135° hooks and extension ≥ 10 d_b; cross-ties/supplementary ties so that every corner bar and alternate bar is supported (no bar > 200 mm from a tied bar).
- Area of hoop: $A_{sh} = 0.18\,s\,h\,(f_{ck}/f_y)(A_g/A_k - 1)$, not less than $0.05\,s\,h\,(f_{ck}/f_y)$; circular spiral: $A_{sh} = 0.09\,s\,D_k\,(f_{ck}/f_y)(A_g/A_k - 1)$; h = longer dimension of rectangular hoop; A_k = core area.
- Remaining column height: spacing ≤ 6 d_b (smallest long. bar) and ≤ 150 mm.
- Strong column - weak beam: $\Sigma M_c \geq 1.4\,\Sigma M_b$ (2016) [1.1 in older code]. Design shear in columns: V_e from column plastic moments $V_e = 1.4\,(M_{u,col}^{top} + M_{u,col}^{bottom})/h_{st}$ (not less than analysis shear).
- Special confining continue through the joint and above/below beam up to l_0 (full height if column is short or below discontinuous walls).

4.3 Beam-column joints

- Joint = a vulnerable region with high shear ($V_{joint} = T_{beam} + C_{beam} - V_{col}$): joint shear failure causes collapse (Northridge, Kobe, Bhuj).
- Confine the joint with closed hoops at spacing ≤ 100 mm (same as column ends) - at least 2 layers of hoops through joint, with ≥ 135° hooks.
- Beam bars anchored in column with development length measured from the inner face of the column core; if column depth < required, bend bars into joint (hook 90° with 8 d_b length in the confined core).
- Interior joint: nominal shear stress limit; exterior joint: anchorage of top & bottom bars with 90° bends into core; corner joints with eccentric beams need larger columns.
- Where beam width > column width, extra hoops required; avoid eccentric beams > b/4.

PART E : BEHAVIOUR OF RC MEMBERS DURING EARTHQ

Member	Behaviour / typical damage	Prevention
Beams	Flexural hinges near column faces (ductile - desirable); cracks, spalling; shear failure (X-cracks) if stirrups sparse; bar slip/anchorage; torsion cracks in edge beams	Under-reinforced, closed stirrups at 2d, capacity shear, 135° hooks
Columns	Hinging at ends (spalling, bar buckling) = storey mechanism; shear failure (X-cracks) in short columns; crushing / axial failure under high axial load; lap-splice failure at base; bending-compression failure in corner columns	Strong column - weak beam, closely spaced hoops over l_0, limit axial load, no lap at ends, avoid short columns
Beam-column joints	Diagonal shear cracks across joint; loss of bond/anchorage (bar pull-out); joint failure -> collapse; exterior and corner joints worst	Joint hoops; proper anchorage; $\Sigma M_c \geq 1.4\,\Sigma M_b$
Slabs/diaphragms	Cracking at openings, re-entrant corners, weak connection to walls; punching shear failure in flat slabs (no beams)	Beam-column frame, trim bars, slab bands
Shear walls	Flexural hinge at base, boundary-element crushing, sliding shear, diagonal tension, coupling-beam failure	Boundary elements, 2 curtains, capacity design
Infill + frame	Infill cracking, short columns, soft storey	Isolated infill, full-height walls
Foundations	Differential settlement, liquefaction, pile failure	Ties, raft/piles, soil improvement

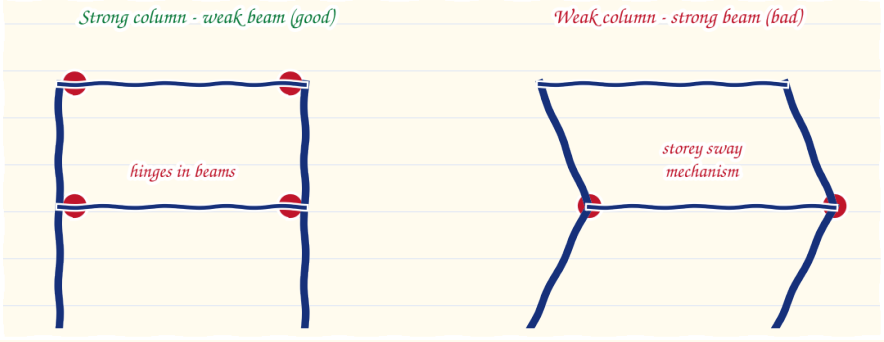


Fig: Desirable beam mechanism vs undesirable storey mechanism

- *Sequence of ideal failure: beam yielding in flexure -> beam hinges form at all floors -> large energy dissipation -> collapse only after large drifts. Avoid: shear failure of beams/columns, joint failure, anchorage failure, column hinging.*
- *Hysteretic behaviour: ductile members show stable fat loops; poorly detailed ones show pinching, strength decay and stiffness degradation after few cycles.*
- *Beams with deep sections and large span-to-depth give more ductility; short deep beams fail in shear.*
- *Columns: ductility drops sharply with high axial load; weak column failure leads to pancaking (Bhuj/Kobe).*

Quick example: Ash for column hoops

Column 400 x 400, cover 40 mm, hoop dia 10 mm -> core 320 x 320 (to hoop centre-line h = 320 mm); A_g = 160 000, $A_k = 320^2$ = 102 400 mm²

f_{ck} = 25, f_y = 415, S = 100 mm: $A_{sh} = 0.18 \times 100 \times 320 \times (25/415) \times (160000/102400 - 1) = 0.18 \times 100 \times 320 \times 0.0602 \times 0.5625$ = 195 mm²

Provide 10 mm hoops with 2 legs each way (2 x 78.5 = 157 mm²) -> add cross-tie / use 3 legs (235 mm²) ✓ or reduce spacing to 75 mm.

PART F : WORKED EXAMPLES

Ex.1 Is a boundary element needed? (shear wall)

Wall l_w = 4.0 m, t = 0.20 m, f_{ck} = 25 MPa; at base (factored): P_u = 3000 kN, M_u = 1200 kN.m

A = 0.8 m²; $Z = t\,l_w^2/6 = 0.2 \times 16/6 = 0.533$ m³

$\sigma_{max} = P_u/A + M_u/Z = 3750 + 2250 = 6000$ kN/m² = 6.0 MPa; $0.2\,f_{ck}$ = 5.0 MPa

6.0 > 5.0 -> boundary elements required at both ends; they can be stopped where $\sigma < 0.15\,f_{ck}$ = 3.75 MPa (up the height).

Ex.2 Beam ductile detailing numbers (IS 13920:2016)

Beam 300 x 500 mm, d = 450 mm, smallest long. bar 16 mm, f_{ck} = 25, f_y = 415

$\rho_{min} = 0.24\sqrt{25/415} = 0.289\%$ -> $A_{st,min} = 0.00289 \times 300 \times 450$ = 390 mm²; ρ_{max} = 2.5% -> 3375 mm²; b/D = 0.6 ≥ 0.3 ✓

Hoop zone length = 2d = 900 mm at each end; spacing = min(d/4 = 112, 8 x 16 = 128, 100) = 100 mm; first hoop at 50 mm -> hoops at 50, 150, ... 850 mm (9 hoops)

Outside the zone: spacing ≤ d/2 = 225 -> provide 200 mm. Hook: 135° with extension ≥ 10 x 8 = 80 mm (≥ 75 mm) for 8 mm hoops.

Ex.3 Joint shear (capacity-based, approx.)

Beam top steel 3 - 20 mm (942 mm²), bottom 3 - 16 mm (603 mm²); steel stress 1.25 f_y = 519 MPa (overstrength)

T_1 = 519 × 942 = 489 kN; C_2 = 519 × 603 = 313 kN; column shear V_{col} ≈ 130 kN

$V_{joint} = T_1 + C_2 - V_{col}$ = 489 + 313 - 130 = 672 kN; joint area 400 × 400 -> τ = 672e3/160000 = 4.2 MPa (compare with code limit ≈ 1.2-1.5 $\sqrt{f_{ck}}$ = 6-7.5 MPa) ✓ - provide joint hoops @ 100 mm.

Ex.4 R from ductility

If displacement ductility available μ = 4: equal displacement rule (T > 0.5 s): R_μ = 4; equal energy (short period): $R_\mu = \sqrt{(2 \times 4 - 1)}$ = 2.65

With over-strength Ω = 1.6 -> $R = R_\mu \Omega$ = 4 × 1.6 = 6.4 (long period); codes adopt conservative R = 5 for SMRF.

COMPARISON & SUMMARY TABLES

Item	Ordinary (non-ductile) detailing	IS 13920:2016 ductile detailing
Stirrups	90° hooks, 150-300 mm spacing	135° hooks, 100 mm in hinge zones, first at 50 mm
Column ends	Normal ties	Special confining hoops over l_0, s ≤ 100 mm
Laps	Anywhere	Only in central half of columns / away from joints, hoops over lap
Joint	No ties or few	Hoops continued through joint
Beam steel	ρ limit 4%	ρ_{max} 2.5%, 2 bars through, top-bottom ratios
$\Sigma M_c / \Sigma M_b$	Not required	≥ 1.4 (1.1 in older version)
Shear design	Analysis shear	Capacity (plastic hinge) shear
Steel	Any Fe 415	Fe 415/500/550 (D) with ductility limits

Element	Key IS 13920:2016 numbers
Beam	b ≥ 200; b/D ≥ 0.3; D ≤ L/4; ρ: 0.24$\sqrt{fck/fy}$ to 2.5%; hoop zone 2d; s = min(d/4, 8φ, 100); first at 50
Column	min 300; b/D ≥ 0.4; ρ 0.8-4%; l_0 = max(D, H/6, 450); s ≤ b/4, 75-100; laps central half
Joint	hoops s ≤ 100 through joint; anchorage from inner core face
Wall	t ≥ 150; ρ ≥ 0.25% each way; 2 curtains if t > 200; boundary element if σ > 0.2 fck
Concrete / steel	M20 min (M25 for H > 15 m in zones III-V); Fe 415/500/550(D), elongation ≥ 14.5%

Capacity design - the idea in 5 steps

1. Choose the desired yielding mechanism (beam hinges + base-of-column hinges).
2. Design the yielding members (beams) for flexure; provide ductile detailing.
3. Compute their probable/overstrength capacities (1.25 f_y).

4. Design all other members (columns, joints, shear in beams, foundations) for forces corresponding to those capacities.
5. Detail: hoops, anchorages, laps outside hinge zones.

QUICK REVISION - Likely questions

1. Strategies for the location of shear walls; sectional shapes; effect on torsion.
2. Cantilever shear wall behaviour; slender vs squat; IS 13920 provisions for walls and boundary elements.
3. Non-structural elements: failure mechanisms, effect on structural behaviour, analysis and prevention of damage.
4. Define ductility; types and relationships; factors affecting ductility; why ductility is needed.
5. Ductile detailing of beams and columns as per IS 13920:2016 (with sketches); special confining reinforcement.
6. Behaviour of beams, columns and joints during earthquakes; beam vs storey mechanism.